



A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State, Challenges, and Future Directions

Foivos Psarommatis^{1,2,3}  · Gokan May⁴

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Abstract

Digital Product Passports (DPPs) are rapidly emerging as core infrastructure for sustainable manufacturing, enabling lifecycle transparency, circular economy practices, and regulatory compliance. This paper presents a comprehensive systematic review of 225 peer-reviewed studies and classifies the field across more than twenty attributes spanning study type, sector, data and information modeling, architecture, technology stack, governance, scalability, and sustainability impact. The analysis of the literature shows that the field is dominated by conceptual and framework papers, with limited empirical validation and fragmented modeling practices; blockchain is the most frequently proposed enabler but raises concerns about scalability, cost, and energy use, pointing toward hybrid architectures in practice. Regulatory alignment, especially within the EU, is accelerating, yet global standardization remains uneven, and formalized data models and modeling languages are inconsistently applied. Sustainability claims are widespread, but robust, comparable indicators and lifecycle assessments are rare, constraining evaluation of real-world impact. We synthesize these insights into a unified DPP lifecycle framework based on information inheritance (from components and processes to products and materials), and outline a research-and-policy agenda that prioritizes (i) harmonized open standards (e.g., AAS, RDF/OWL, sector schemas), (ii) hybrid, interoperable technical architectures, (iii) measurable impact metrics and verification, and (iv) inclusive, multi-stakeholder governance with clear roles and incentives. Advancing along these directions can transition DPPs from policy-driven pilots to scalable, auditable infrastructures for digital traceability, resource efficiency, and sustainable industrial transformation.

Keywords Digital Product Passports · Sustainable Manufacturing · Circular Economy · Traceability · Blockchain · Lifecycle Management

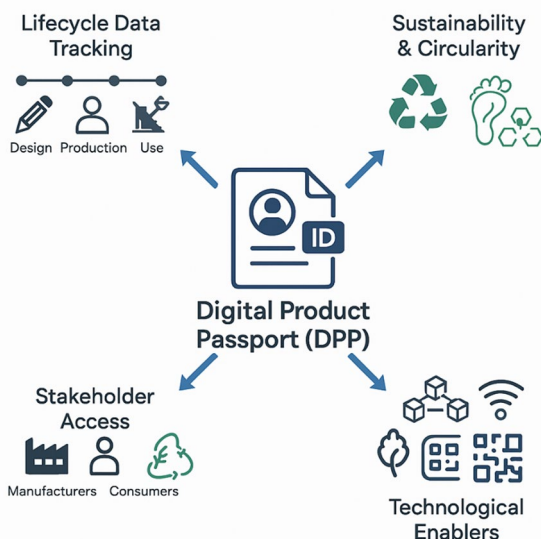
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Introduction

The Digital Product Passport (DPP) is defined as a standardized digital record that travels with a product across its lifecycle, providing detailed information on origin, materials, reparability, reuse, recycling, and overall sustainability attributes [1]. As a key enabler of circular economy practices, the DPP enhances transparency and data sharing among stakeholders, thereby supporting sustainability in modern manufacturing. With industries placing a greater emphasis on sustainability, traceability, and transparency, DPPs offer an alternative that fits within the context of the circular economy and allow for the abandonment of conventional linear production methods. DPPs assist global sustainability goals by facilitating responsible production and material reuse by integrating comprehensive information about a product's materials, environmental impact, and lifetime data into a digital format.

DPPs have been proposed as essential instruments for enabling lifecycle data traceability, regulatory compliance, and circular product management. By aggregating and exchanging structured product information throughout a product's lifecycle from design to end-of-life, DPPs promise to operationalize transparency, sustainability, and informed decision-making across global supply chains. Figure 1 provides a conceptual overview of the core functions of DPPs, highlighting how they integrate technological infrastructure, stakeholder interaction, and lifecycle intelligence to support sustainable manufacturing and circular economy objectives. The capacity to monitor products from the point of production to end-of-life is essential in sustainable manufacturing. DPPs contribute significantly to this by minimizing waste, improving recycling procedures, and guaranteeing resource efficiency and material recovery [2]. DPPs make it possible for stakeholders, including producers, customers, and recyclers, remanufacturers, and reuse actors to get real-time updates on a product's status, including information on its origin, usage, and recyclability. Making educated judgments, maximizing material flows, and enhancing resource recovery all depend on having

Fig. 1 Conceptual overview of the role of DPPs in sustainable manufacturing, illustrating their core functions: lifecycle data tracking, support for circularity, stakeholder access, and technological enablers



access to such thorough data [3]. Emerging technologies such as blockchain, the Internet of Things (IoT), and artificial intelligence (AI) have been shown to support DPP deployment by ensuring data security, enabling real-time monitoring, and offering predictive analytics for decision-making [4–6]. Applications in the automotive sector [7] and electronics [8] illustrate how DPPs enable tracking of critical raw materials, support recycling pathways, and facilitate sustainable product lifecycle management.

Despite their promise, DPP adoption faces significant challenges. One major issue is the lack of industry-wide standards, which complicates framework development and interoperability across global supply chains [9]. Data privacy and intellectual property concerns further hinder information-sharing, especially in competitive industries like automotive and electronics [5, 7]. Cost is another barrier, as the infrastructure needed for IoT, blockchain, and data analytics may be prohibitive for SMEs [10]. Finally, while regulatory initiatives such as the EU's Ecodesign for Sustainable Products Regulation provide momentum [9], global coordination remains limited, creating fragmented standards and compliance frameworks [5].

This article provides an in-depth examination of DPPs and their role in advancing sustainable manufacturing. By synthesizing the most recent research and addressing technological, regulatory, and industrial dimensions, this study closes identified gaps in standardization, regulatory integration, and sector-specific applications.

In contrast to prior research, this review focuses exclusively on the integration of DPPs into sustainable manufacturing methods. While existing literature often centers on technical aspects such as blockchain or data security, this paper extends the discussion to include regulatory frameworks and industry-specific adoption barriers. Despite the growing number of review papers on Digital Product Passports, several systematic gaps remain insufficiently addressed. First, most existing reviews focus on specific technological enablers (e.g., blockchain, IoT, or traceability platforms) or single sectors (e.g., batteries, textiles, construction), without offering a cross-sectoral, multi-dimensional classification of DPP research. Second, prior reviews rarely apply a consistent analytical taxonomy spanning study type, data and information modeling, architecture, scalability, governance, and sustainability impact, which limits comparability across studies. Third, while sustainability is frequently cited as a motivation, existing reviews seldom assess how sustainability impact is defined, evidenced, or measured, leading to largely qualitative or aspirational conclusions. Fourth, the role of formal modeling languages, data standards, and information inheritance across lifecycle stages remains fragmented or overlooked, despite being critical for interoperability and scalability. Finally, most reviews stop at descriptive synthesis and do not consolidate findings into a unified lifecycle-oriented framework that explicitly links DPP design choices to circular economy and industrial implementation needs.

In response to these gaps, this review pursues four explicit objectives:

- (1) to provide a **comprehensive, cross-sectoral synthesis** of DPP research based on a structured taxonomy applied consistently across 225 peer-reviewed studies.
- (2) to analyze DPP implementations across **technical, informational, organizational, and regulatory dimensions**, including study type, data and information modeling, architecture, scalability, governance, and stakeholder involvement.
- (3) to critically assess **how sustainability impact is conceptualized and evidenced** in the literature, distinguishing between high-, medium-, and partially substantiated claims.

(4) to propose a **holistic DPP lifecycle framework** grounded in the principle of information inheritance, spanning raw materials, manufacturing, use, maintenance, remanufacturing, and recycling.

Research Questions

Given the exploratory and integrative nature of this systematic literature review, the study is guided by a set of research questions rather than formal hypotheses. The objective is to synthesize, classify, and critically assess the existing body of knowledge on Digital Product Passports in sustainable manufacturing. Accordingly, the review addresses the following research questions:

RQ1: How is existing DPP research distributed across study types, industrial sectors, and product categories?

RQ2: What technological architectures, data and information modeling approaches, and digital infrastructures are proposed or applied in DPP implementations?

RQ3: How are governance models, stakeholder roles, and regulatory alignment addressed in the DPP literature?

RQ4: How is sustainability impact conceptualized, evidenced, and assessed across existing DPP studies?

RQ5: To what extent do current studies support a lifecycle-oriented and interoperable DPP framework spanning design, manufacturing, use, and end-of-life stages?

The remainder of this paper is structured as follows: Sect. "[Previous Review Papers on DPP](#)" reviews previous literature reviews on DPPs and their limitations. Sect. "[Methodology](#)" outlines the methodology used for systematic paper selection and coding. Sect. "[Results](#)" presents the results across multiple analytical dimensions. Sect. "[DPP life cycle Framework](#)" develops a comprehensive framework for the DPP life cycle, detailing its role from raw material extraction through manufacturing, use, maintenance, remanufacturing, and recycling. Sect. "Critical Discussion" offers a critical discussion of systemic challenges and opportunities, and Sect. "Conclusions" concludes with actionable recommendations for research, policy, and industry.

Previous Review Papers on DPP

To identify existing review studies on DPPs, we conducted a structured search across Scopus, Web of Science, and Google Scholar. The search strings combined the terms "Digital Product Passport" with "review," "survey," or "literature review." We applied the following inclusion criteria: (i) the paper explicitly focuses on DPP or closely related concepts in circular economy and sustainable manufacturing; (ii) the study is peer-reviewed and published in English; and (iii) it provides a structured review, survey, or synthesis of prior literature. Papers that only mention DPP in passing without providing a substantive review were excluded. This process initially yielded 34 records, of which 19 met all inclusion criteria after screening titles, abstracts, and full texts. These 19 papers represent previous review

studies and are distinct from the main corpus of 225 primary research articles that are systematically analyzed in Sect. “Methodology”.

The DPP idea has quickly become well-known in conversations about sustainability and the circular economy. A wide range of frameworks, techniques, and applications are reflected in the growing body of literature that aims to use DPPs to improve data interchange, traceability, and transparency throughout product lifecycles. This section synthesizes contributions from 19 key papers, providing a comprehensive view of the state of research on DPPs and their role in supporting sustainable manufacturing and circular economies. Below is the timeline showing how the number of DPP review papers has evolved from 2017 to 2024 (Fig. 2). It illustrates a clear acceleration in scholarly interest, after 2020, reflecting both the growing maturity of the concept and strong policy drivers. Early studies between 2017 and 2019 were sparse and exploratory, but since 2021 the number of reviews has expanded significantly, coinciding with the launch of the European Green Deal and the Circular Economy Action Plan, and most recently the Ecodesign for Sustainable Products Regulation (ESPR) [9, 10]. These frameworks positioned DPPs as a cornerstone for sustainable production, material traceability, and compliance, creating strong incentives for researchers to explore technical, regulatory, and governance aspects of the topic. Recent reviews confirm this shift, showing a diversification of themes, from blockchain and IoT integration to sector-specific applications in fashion, batteries, and construction [5, 11, 12] indicating that DPPs have moved from a niche idea into a mainstream sustainability research domain.

This growth in scholarly output is also reinforced by industrial and market developments. The global DPP market was valued at over USD 200 million in 2024 and is projected to expand at an annual growth rate exceeding 30% by 2030 [13], underlining the practical relevance of the field. Industry pilots in textiles, electronics, and batteries demonstrate increasing adoption, with leading fashion brands and automotive manufacturers testing DPPs for transparency, circularity, and compliance [8, 14]. At the same time, initiatives such as the European Battery Passport and textile-focused pilots highlight how regulation is being operationalized through real-world platforms [3, 15]. Together, these academic, regulatory, and industrial drivers explain the sharp rise in review literature captured in Fig. 2, and they signal that DPP research is entering a phase of consolidation and empirical validation.

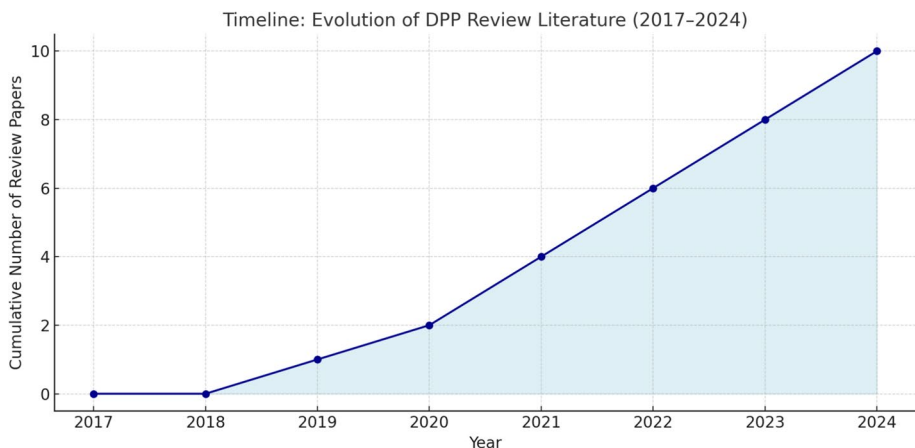


Fig. 2 Evolution of DPP Review Literature (2017–2024)

Psarommatis and May [16] highlight the potential of DPPs in modern manufacturing by enhancing transparency and sustainability [1]. Their paper focuses on how DPPs facilitate improved information flow across product lifecycles, fostering circularity in production processes. This perspective is echoed by Hariyani et al. [11], who emphasize the role of digital technologies in advancing circular economy practices and how DPPs are integral to enhancing lifecycle analysis [11]. Both papers underscore the central role of DPPs in bridging information gaps that inhibit circularity.

DPPs' architectural considerations are explored by Wiesner et al. [12], who propose a reference architecture that supports batch-level traceability in manufacturing supply chains [12]. This is further complemented by Greiner et al. [5], who investigate blockchain-based consortia to improve interoperability between DPP systems, demonstrating the use of distributed ledgers to secure and share sustainability-related information [5].

In supply chain-specific studies, Chaudhuri et al. [4] explore DPP applications in electronics, presenting pathways for embedding circularity through transparent data exchange [4]. Zhang and Seuring [2] extend this discussion by reviewing various use cases of DPPs in sustainable supply chain management, emphasizing the role of digital tools in facilitating circular supply chains [2].

The role of blockchain and decentralized systems is reinforced by [5], who focus on secure, decentralized data exchange in DPP systems, a critical factor for ensuring transparency in circular economy ecosystems. Similarly, Voulgaridis et al. [6] provide a broader technological framework for DPPs as enablers of the digital circular economy, particularly through the integration of IoT and data exchange protocols [6].

Jakubowicz and Yarahmadi [8] delve into traceability techniques, emphasizing how DPPs can manage ABS plastics and improve recycling processes [8]. Rumetshofer and Fischer [15] build on this by focusing on plastic material tracking, presenting DPPs as a critical tool for material traceability, which is essential for the circular economy [15].

An earlier investigation by Walden et al. [17] frames DPPs as key enablers of the circular economy, focusing on their role in product lifecycle management and transparency [17]. This view is shared by [18], who identify the specific data needs for integrating DPPs into lifecycle decision-making, providing detailed insights into how DPPs support sustainable production and consumption [19].

The practical challenges of DPP implementation are addressed by [20], who highlight the potential for DPPs to enhance data-driven lifecycle analysis. In contrast, [21] propose a universal definition for DPP ecosystems, focusing on the stakeholder concerns that must be addressed to achieve widespread adoption.

Koppelaar et al. [3] extend the application of DPPs to critical raw materials, specifically focusing on reuse and recycling [3]. Their study underscores the growing need for DPPs in mitigating resource scarcity, particularly in the context of global supply chain disruptions.

In terms of technical and legal requirements, [22] investigate how DPPs must align with existing circular economy regulations, ensuring that the systems not only support technical interoperability but also meet legal and sustainability standards [10]. Jansen et al. [9] provide a broad overview of ongoing initiatives, outlining various projects that contribute to the development and deployment of DPP systems across different sectors [9].

Building on this, Jansen et al. [23] identify the core technical requirements for DPP systems, derived from structured stakeholder analysis, offering insights into what needs to be addressed for successful DPP deployment [23]. Finally, [24] focus on the adoption of

DPPs along the electric vehicle battery (EVB) value chain, discussing enablers and barriers to adoption, thereby contributing to the understanding of sector-specific challenges in DPP implementation [25].

Summary Table: Contributions from Literature

Theme	Key Contributions from Literature	Relevance to Current Study
DPP in Circular Manufacturing	[11, 16] discuss DPPs' role in enabling circularity by improving transparency and lifecycle analysis	Sets the foundation for understanding DPP's strategic role in sustainability
DPP Architecture & Interoperability	[5, 12] propose blockchain and batch-level reference architectures for secure and interoperable DPP systems	Crucial for developing a robust, scalable DPP system architecture
Sector-Specific DPP Applications	[2, 4] explore DPP use cases in electronics and supply chain management	Illustrates sector-specific DPP applications, particularly in electronics and supply chains
Blockchain & Decentralization in DPPs	[6, 26] emphasize blockchain and IoT frameworks for decentralized DPP systems	Demonstrates the technological enablers of secure, decentralized DPP systems
Traceability & Recycling	[8, 15] focus on material traceability and recycling using DPPs	Key for understanding how DPPs can improve traceability and recycling processes
Lifecycle Management & Data Needs	[17, 19] emphasize lifecycle management and data requirements for DPP integration	Provides insights into the data needs for integrating DPPs into lifecycle management systems
Challenges in DPP Adoption	[20, 21] discuss the challenges of DPP adoption and propose a universal definition for DPP ecosystems	Identifies challenges and adoption pathways critical for ensuring widespread use of DPPs
DPP in Resource Management	[3] focus on DPPs' role in critical raw material reuse and recycling	Highlights the importance of DPPs in managing scarce resources
Technical & Legal Requirements	[9, 10, 27] explore technical and legal considerations for DPP implementation	Ensures the alignment of DPP systems with technical standards and legal frameworks
Adoption Barriers in DPPs	[23, 24] focus on adoption barriers and stakeholder requirements, particularly in the EVB sector	Sector-specific insights into adoption challenges, particularly for EVB applications

From the review of these 19 papers, DPPs are a cornerstone in advancing circular economies and sustainable manufacturing practices. The variety of strategies, from traceability systems for recycling to blockchain-based architectures, illustrates the wide applicability of DPPs across sectors and businesses. Important obstacles still need to be overcome, mainly in the areas of stakeholder engagement, data governance, and interoperability.

Many people support the use of blockchain for safe, decentralized data exchange [5, 26], although [10] point out that concerns about data privacy and regulatory compliance require more research. Furthermore, research indicates that although technical criteria for DPPs are becoming better understood, more precise legal frameworks are still required to guarantee their widespread adoption [20, 21].

Finally, adoption barriers, particularly in sector-specific contexts such as the electric vehicle battery value chain [7, 24, 25], underscore the importance of aligning DPP system design with the specific needs of industries and stakeholders. Without overcoming these barriers, the full potential of DPPs to foster circularity may not be realized. Future research should focus on addressing these sector-specific challenges while maintaining the flexibility to scale DPP systems across multiple industries.

While existing review papers have significantly advanced understanding of DPP concepts, technologies, and sectoral applications, they collectively exhibit several limitations. Most reviews focus on isolated dimensions, such as blockchain architectures, regulatory drivers, or sector-specific pilots, without integrating technical, informational, and governance aspects into a unified analytical framework. Moreover, few reviews systematically examine data and information modeling practices, despite their centrality to interoperability and lifecycle continuity. Sustainability impacts are often asserted but rarely categorized or evaluated in a comparable manner. Finally, none of the reviewed studies provides a lifecycle-spanning framework that operationalizes DPPs as dynamic, evolving information artifacts across multiple product lives. These limitations motivate the comprehensive, taxonomy-driven and lifecycle-oriented approach adopted in this study.

Methodology

We define our structured literature review (SLR) methodology in this section for DPPs in sustainable manufacturing. A methodical and replicable approach was employed to gather a thorough, pertinent, and high-caliber dataset to examine the present condition, challenges, and future trajectories of DPPs.

We employed a systematic literature review to consolidate existing knowledge. This approach was selected to identify essential themes, knowledge deficiencies, and trends while ensuring transparency and replicability. The SLR directs DPP research analysis across various industries, emphasizing its applications, technological facilitators, and obstacles. The review employed a predetermined analytical framework to classify studies based on DPP objectives, technological architecture, and sustainability effects. The study ensured quality and relevance by using rigorous inclusion and exclusion criteria.

By using a structured review approach, this work synthesizes the present collection of knowledge systematically and comprehensively. This method was selected to identify main themes, knowledge gaps, and trends while remaining transparent and reproducible. The SLR is the basis for examining DPP research across diverse fields, focusing on applications, technological benefits and challenges. Studies were organized according to their long-term sustainability benefits, technological enablers, and stated objectives. Rigorous guidelines for inclusion and exclusion ensured that only relevant, peer-reviewed, and substantive works were retained.

Using well-known academic databases ScienceDirect, Scopus, Web of Science (WoS), and IEEE Xplore, we conducted our literature search. These databases were chosen as they cover a wide range of disciplines, particularly in new technologies, manufacturing, and sustainability. The search employed combinations of the terms “Digital Product Passport” OR “DPP” with “sustainability”, “circular economy”, and “traceability” using Boolean operators to refine results. The initial search yielded 157 ScienceDirect papers, 101 Scopus papers, 38 Web of Science papers, and 10 IEEE Xplore papers, for a total of 306 records. After removing 49 duplicates, the dataset comprised 257 unique records.

The screening process applied predefined criteria:

Inclusion and Exclusion Criteria

The selection of studies followed explicitly defined inclusion and exclusion criteria to ensure relevance, quality, and consistency.

Inclusion criteria:

- (i) peer-reviewed journal articles, conference proceedings, or book chapters
- (ii) publications written in English
- (iii) studies that explicitly address Digital Product Passports or closely related product life-cycle data mechanisms within the context of sustainability, circular economy, traceability, or regulatory compliance.

Exclusion criteria:

- (i) non-peer-reviewed sources such as reports, white papers, theses, or policy briefs
- (ii) publications that mention DPPs only superficially without substantive conceptual, technical, or analytical contribution
- (iii) duplicate records across databases.

In addition, a limited number of studies in which DPPs were not the primary focus but were treated as part of broader discussions on digital traceability, lifecycle data management, or circular business models were retained when they provided meaningful contextual or methodological insights. Studies that only referenced DPPs tangentially, without analytical relevance, were excluded. The screening process, including identification, duplication removal, eligibility assessment, and final inclusion ($n=225$), is documented using a PRISMA flow diagram (Fig. 3) [28, 29].

After title, abstract, and full-text screening, 32 records were excluded, leaving a final set of 225 papers included in the review. In applying the criteria, we retained a small number of studies where DPP was not the central topic but was treated as part of a broader discussion (e.g., digital traceability, circular business models), as these provide useful contextual insights. By contrast, papers that only mentioned DPP in passing without substantive relevance were excluded.

The search and selection process is summarized in Fig. 3, which follows the PRISMA guidelines. The diagram illustrates the number of records identified, screened, excluded, and finally included in the review ($n=225$), ensuring transparency and consistency.

An analytical framework stressing important features of DPP research guided the analysis of the selected papers. The characteristics were categorized into general topics covering study type, industry uses, technological framework, challenges, and environmental impacts. Studies were evaluated, for example, depending on their analysis of conceptual frameworks, empirical data, or pilot projects. The framework also documented aspects of technological integration, including blockchain, IoT, and AI, along with the challenges related to the implementation of DPP systems. The data extraction process entailed meticulous examination and coding of the abstract, methodology, and findings sections of each study. This approach assured the methodical gathering of all relevant information, therefore enabling an elaborate study of the body of work. Not explicitly stated attributes were classified as "not stated," coupled with brief clarifying notes.

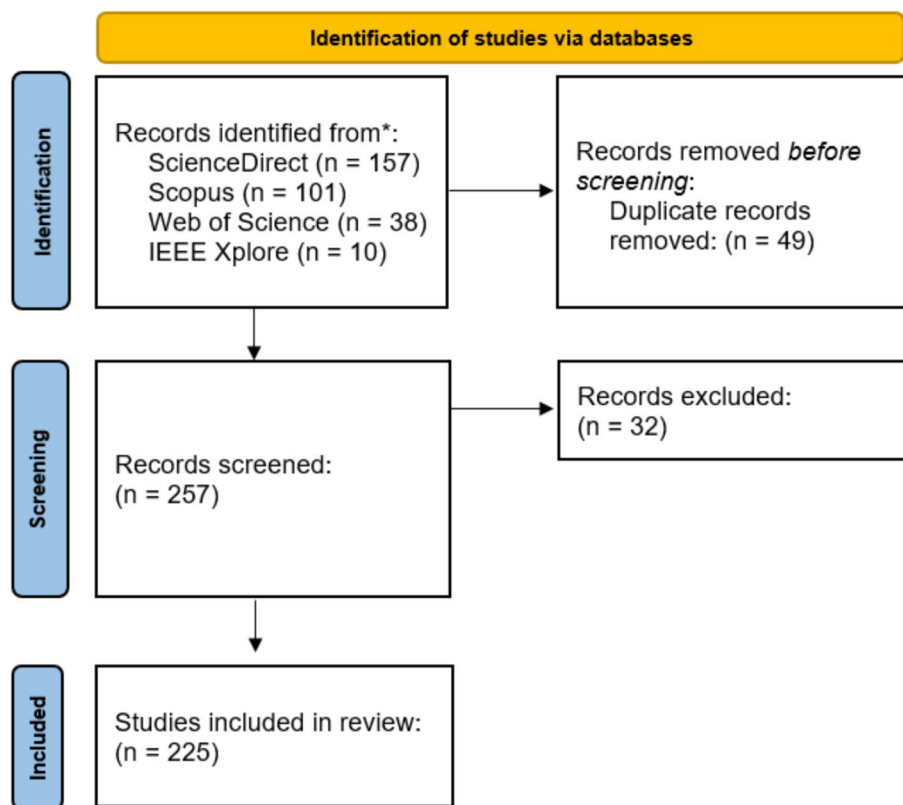


Fig. 3 PRISMA flow diagram

The gathered data were arranged and examined using content analysis methods. Furthermore, the data was administered and represented through Excel, enabling the methodical extraction and classification of information.

The data synthesis process entailed a thematic analysis of the studies, highlighting similarities and disparities in their methodologies, results, and implications. The analysis encompassed attributes including the objectives of DPPs (e.g., traceability, lifecycle management, regulatory compliance), technological frameworks (e.g., centralized, decentralized, hybrid), and challenges (e.g., cost, data privacy, interoperability) to furnish a comprehensive understanding of the domain. We explored main research trends and shortcomings by matching our observations.

Given the extensive scope of this review, covering 225 papers, the handling of references required a balance between completeness and readability. To avoid overwhelming the main text with an excessively long sequence of citations, the decision was made to consolidate the full list of reviewed papers in Annex 1, where they are presented with their detailed classifications. Within the article itself, references are selectively included to illustrate sectoral applications, methodological approaches, or representative examples of high-, medium-, or not-stated sustainability impact. This ensures that the discussion remains narratively clear while still anchored in the evidence base. We recognize that citing all 225 papers in-line

would reduce readability and risk weakening the analytical insights of the review. Instead, our approach preserves transparency (through Annex 1) while allowing the body text to focus on patterns, trends, and representative cases. This strategy is consistent with recent large-sample reviews in sustainability and digitalization research, where annex-based reference reporting is used to manage scale without compromising scientific rigor.

Results

Based on the reviewed literature, we provide a comprehensive overview of the current state of research on DPPs, covering a period of 4 years from 2021 to 2024. In this overview, we cover a wide range of industries, product types, and use cases. In this section, the quantitative findings of the dataset analysis are presented, with the number of papers for different classes of each attribute as well as percentage distributions as supporting evidence of the findings. Accordingly, we discuss each attribute in terms of its dominance, trends across domains, and implications for research and practice. Figure 4 illustrates distribution of papers distribution per year. The figure shows an exponential growth in the number of the papers related to DPP.

It should be noted that in some of the following results the category “Other” aggregates cases with only one or two occurrences in the dataset. Because these values appear too infrequently to allow for meaningful separate categories, they were grouped together to maintain readability of the figures. Illustrative examples of such cases are mentioned in the text where relevant.

Study Types

Based on the results presented in Fig. 5, the majority of the papers we analyzed employ a conceptual or framework-based methodology, and are in the preliminary and exploratory phase of research on DPPs. 95 papers (42.22%) were classified as conceptual studies, and focus on overarching ideas, preliminary frameworks, and alignment with policy agendas instead of presenting applied findings. These contributions frequently describe the characteristics of DPPs, their governance structures, and their potential to facilitate sustainability or transitions to a circular economy. Although ambitious, many of these papers lack implementation strategies and validation steps, and concentrate instead on system level analysis.

Fig. 4 Year distribution of papers

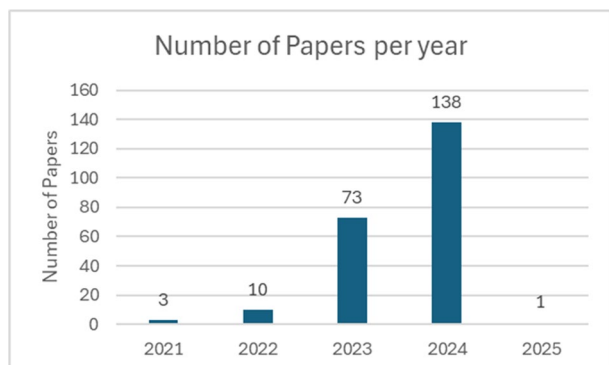
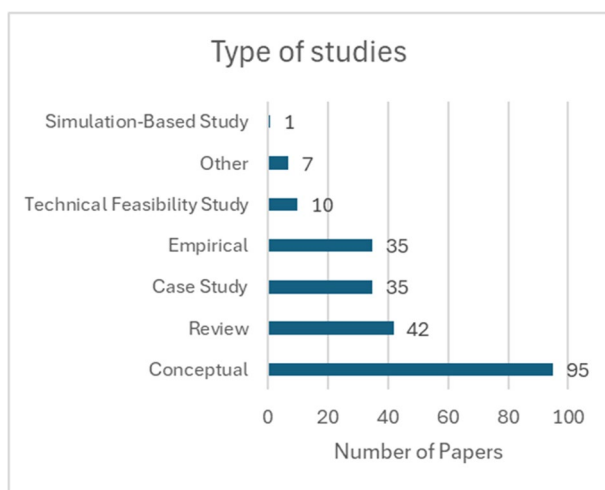


Fig. 5 Type of Studies

Review articles represent the second largest category with 42 papers (18.67%) that offer structured summaries of existing research. These studies consolidate definitions, evaluate policy strategies, and define future research requirements. Most are narrative reviews without a formal methodology, although a few adopt structured or thematic approaches to classify existing DPP initiatives and use cases.

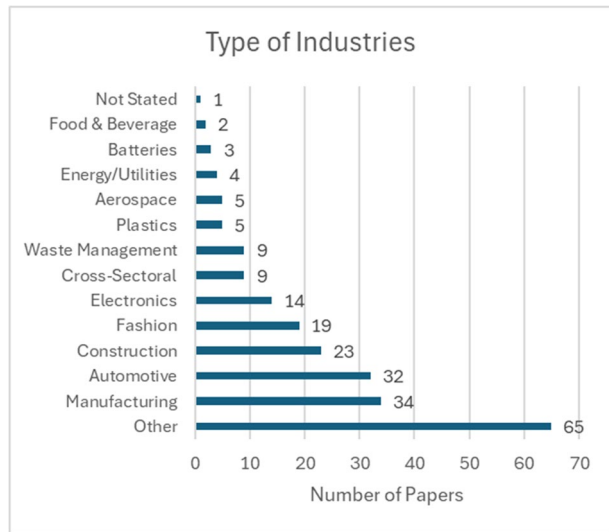
Application focused research is emerging, but still less common. 35 papers (15.56%) were classified as case studies, typically describing DPPs in action within specific contexts such as fashion, batteries, or electronics. These studies typically illustrate feasibility or describe real world barriers, yet they demonstrate considerable variability in depth and detail. In the same vein, 35 papers were classified as empirical studies (15.56%), and utilize data from stakeholders, surveys, and interviews, among others. Some of these are pilot implementations, while others concentrate on perceptions, preparedness, or organizational effects.

10 articles (4.44%) in our analysis focus on the practicality of DPP related enabling technologies such as blockchain, the IoT, and artificial intelligence. Focused studies like these often use data architecture diagrams, performance evaluations, and prototype platforms. Despite being rare, the few that do exist provide insight on important design trade-offs and practical constraints.

Based on the breakdown of study types, most DPP research is conceptual. However, there is an increasing trend towards empirical and case-based studies, that shows implementation and validation are becoming more important. The significant presence of reviews indicates a collective endeavor to integrate emerging knowledge and situate DPPs within the wider contexts of digital transformation and sustainability discussions.

Applications in Industry and Products

The examination of industry applications indicates that the majority of papers employ a cross-sectoral approach, with 65 papers (28.89%) categorized as addressing multiple or general industry contexts (Fig. 6). These studies typically seek to establish comprehensive frameworks, infrastructure models, or policy structures applicable across various industries.

Fig. 6 Types of industries

Instead of concentrating on a specific sector, they highlight the importance of flexibility, interoperability, and overarching design principles for DPP systems.

The manufacturing sector is the most extensively researched, with 34 papers (15.11%) focusing on discrete or process industries. These studies examine how DPPs can enhance product lifecycle transparency in smart factory environments and improve supply chain traceability. For example, [30] propose a framework for classifying product data in manufacturing to support DPP development, while [12] outline a reference architecture enabling batch-level traceability across manufacturing supply chains. In this classification, the category “manufacturing” refers to studies that address DPPs in general industrial or production settings without focusing on a specific sectoral domain. These papers typically discuss frameworks, architectures, or cross-cutting applications applicable across discrete or process industries. By contrast, papers explicitly addressing particular domains such as automotive or electronics were coded separately to avoid double counting. This distinction ensures that “manufacturing” captures only those contributions with a broad or non-sector-specific industrial orientation, while sector-specific applications are reflected in their dedicated categories.

The automotive sector follows closely, with 32 papers (14.22%). A large portion of these address electric vehicle batteries, reflecting both industrial momentum and the EU Battery Regulation [25], analyze enablers and barriers for DPP adoption along the EV battery value chain, while [3] emphasize critical raw material traceability for enabling reuse and recycling. Together, these contributions highlight automotive batteries as a testbed for regulatory and technical experimentation.

The construction sector is covered in 23 papers (10.22%), where DPPs often take the form of material passports. These papers focus on lifecycle design, carbon accounting, and the integration of Building Information Modeling (BIM). For instance, [31] identify barriers and opportunities for DPP implementation in construction projects, while [32] propose a modular ontology for developing DPPs to promote circularity in the built environment.

Fashion and textiles account for 19 papers (8.44%), primarily focused on supply chain traceability and consumer-facing transparency. Alves et al. [14] introduce a traceability platform that uses DPPs to monitor environmental and social sustainability in clothing production [14], demonstrating how product-level information can improve accountability across suppliers. Jæger and Myrold [33] present a consumer-oriented pilot where DPPs are used to provide end-users with accessible information on product origin and sustainability credentials, highlighting the potential for strengthening consumer trust [33]. Similarly, Spiß et al. [34] explore the integration of DPPs into apparel supply chains, focusing on interoperability challenges and the alignment of digital infrastructures [34]. Together, these studies illustrate how the textile sector is positioning DPPs not only as compliance tools under the ESPR but also as mechanisms to engage consumers and close information gaps in global value chains.

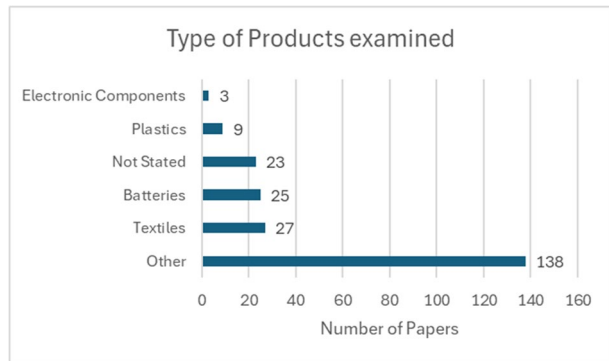
Smaller but growing clusters address energy (15 papers; 6.67%), electronics (10 papers; 4.44%), food and beverage (5 papers; 2.22%), and aerospace (5 papers; 2.22%). In plastics and electronics, [8] review traceability mechanisms applicable to ABS plastics [8], while Rumetshofer and Fischer [15] explore DPPs for tracking recycled plastics [15]. In food systems, Hoang and Böckel [35] show how DPPs can integrate carbon footprint metrics in agri-food supply chains [35]. In aerospace, [36, 37] analyze DPP requirements for e-waste disassembly and circular management of critical components [36].

At the product level, the most common category remains papers without a product-specific focus (138 papers; 61.33%), instead discussing DPPs as cross-cutting frameworks. Among product-focused contributions, textiles appear in 27 papers, highlighting traceability and recyclability of fibers, while batteries are discussed in 25 papers, mainly linked to EV applications and raw material recovery. Plastics are addressed in 9 papers, primarily emphasizing verification of recycled content [15]. Smaller numbers of papers examine electronics and food and beverage products, suggesting that research is still at an early stage but expanding.

While the previous subsection highlighted DPP applications at the industry and value chain level, a smaller but significant share of studies address specific products or materials, focusing on traceability, labeling, and recyclability. The largest group of papers, however, does not concentrate on a single product: 138 studies (61.33%) discuss DPPs in cross-sectoral terms, proposing architectures, governance frameworks, or data models applicable across product categories. These include, for example, [12], who develop a reference architecture for batch-level traceability, and [1], who situate DPPs as key enablers of circularity in remanufacturing contexts. The “Other” category also includes papers addressing niche or highly diverse product types with very low frequency, such as construction materials (e.g., concrete, insulation, prefabricated modules), footwear, electric bike motors, metal-cutting tools, dairy products, photovoltaic modules, or critical raw materials. These were grouped together to avoid excessive fragmentation, as each appears in only one or two studies.

Among the product-specific studies (Fig. 7), textiles stand out with 27 papers, reflecting their priority status under the ESPR. Here, the emphasis is on fiber-level traceability and consumer-facing labeling. Alves et al. [14] propose a DPP-based platform for monitoring sustainability attributes across clothing value chains [14], while Jæger and Myrold [33] show how product-level data can be embedded into garment passports to inform consumers directly [33]. These works highlight how DPPs move beyond supply chain management to influence product labeling and responsible consumption.

Fig. 7 Products examined in the papers



Batteries are the next most studied category (25 papers), frequently linked to electric vehicles but framed specifically around raw material recovery and end-of-life management. Koppelaar et al. [3] focus on tracking critical raw materials within battery passports, enabling reuse and recycling strategies [3], while [7, 24, 25] examine adoption barriers at the value chain scale. These studies show that battery-focused DPPs often act as prototypes for material-specific traceability [25].

In plastics (9 papers), the emphasis is on verification of recycled content and polymer-specific recovery pathways. Rumetshofer and Fischer [15] present approaches for monitoring flows of recycled plastics to ensure compliance and transparency [8, 15] analyze how DPPs could be integrated into polymer recycling to enhance traceability.

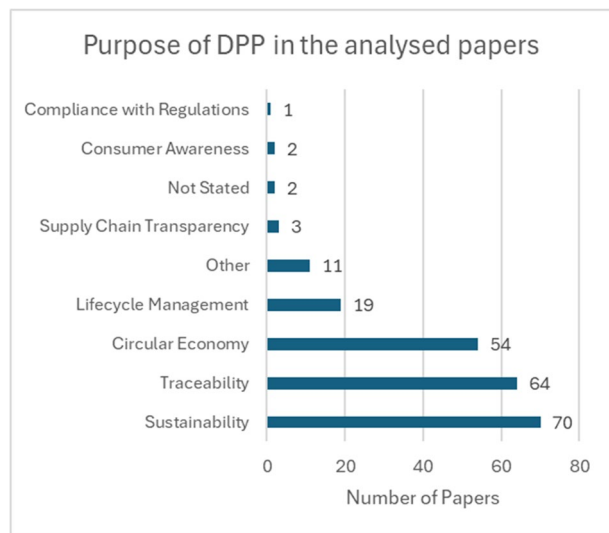
A smaller set of papers address electronics and food and beverages. For instance, Hoang and Böckel [35] discuss integrating carbon footprint metrics into agri-food passports, while [36, 37] examine requirements for electronic component passports in e-waste disassembly. Although these sectors are underrepresented (fewer than 10 studies each), they indicate the breadth of emerging product-level applications.

Finally, 23 papers (10.22%) mention product implications indirectly, linking DPP frameworks to categories such as construction materials or machinery without focusing on a single product. These contributions demonstrate that while most DPP research still operates at the system or industry level, concrete product-centered studies are growing, emphasizing item-level traceability, recyclability, and labeling.

Purpose and Functional Orientation

The literature review of our study indicates that DPPs primarily serve to facilitate transitions to a circular economy, improve sustainability, and strengthen product traceability (Fig. 8). Sustainability emerges as the most frequently referenced goal, mentioned in 70 papers. These studies demonstrate that DPPs function as tools that facilitate the achievement of environmental objectives, including lifecycle impact assessment, eco-design, carbon tracking, and sustainable sourcing. Many studies highlight the strategic importance of DPPs as essential information foundations for future regulatory compliance, and associate them with policy instruments like the Corporate Sustainability Reporting Directive and the European Green Deal.

Traceability is the main theme in 64 papers, and this reflects widespread interest in DPPs as tools for tracking material origin, ownership, and processing across global supply chains.

Fig. 8 Purpose of the DPPs

These papers often highlight the value of DPPs in increasing supply chain transparency, reducing fraud, and enabling more responsible consumption.

There are 54 papers that discuss the functionalities of the circular economy. These papers illustrate DPPs as facilitators of product reuse, remanufacturing, recycling, and reverse logistics. These studies emphasize end-of-life management and use DPPs to facilitate design-for-disassembly, component separation, or automated sorting. The construction and automotive industries are among the most active in this sector.

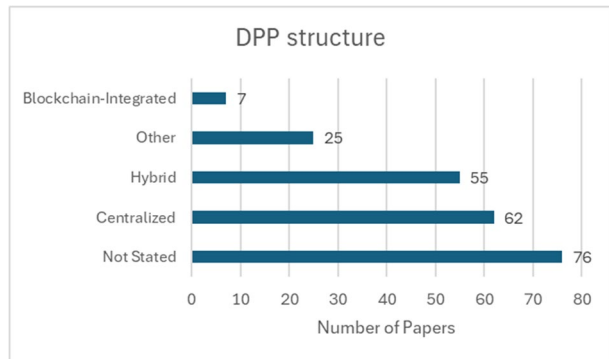
Lifecycle management is the main focus in 19 papers. These studies describe DPPs as longitudinal data containers that track a product through design, production, use, maintenance, resale, and disposal. Applications often involve smart products, service-based business models, or high-value assets where performance data and usage history are relevant to future owners or service providers.

11 papers in our selection highlight new uses for DPPs. Among other features, these applications include enhancing customer engagement, providing assistance for predictive maintenance, and providing data for policy analytics. This demonstrates how the DPP is becoming increasingly active in areas beyond typical compliance and reporting requirements.

DPP Structures

The structural design of DPP systems differs significantly across the literature, highlighting different architectural assumptions, data management strategies, and stakeholder responsibilities (Fig. 9). The majority of studies (92 papers) focus on multi-layered DPP designs, which typically include levels for data collecting, storage, access management, and user interfaces. These models stress modularity and flexibility, allowing the DPP to adapt to a variety of data types, lifecycle phases, and stakeholders' views. These structures are typically proposed in research that focus on cross-sectoral or framework-oriented methods, with the aim of gaining broad application and scalability.

Fig. 9 DPP structure from the analyzed papers



51 papers discuss centralized models, frequently focusing on oversight and regulatory compliance as key issues. These frameworks suggest that a central authority or registry should oversee DPP data and the rights to access it. Research on regulated industries, especially in automotive and construction, often favors centralized models, highlighting benefits such as auditability, consistent standards, and legal accountability.

Decentralized architectures are discussed in 37 publications, several of which utilize blockchain or distributed ledger technology to manage trust, authenticity, and data ownership. These models are becoming more common in studies concerning textiles, electronics, and multi actor value chains, especially in contexts where power disparities or cross-border issues make centralized governance unfeasible. Authors often highlight the benefits of decentralization, like openness, resistance to interference, and peer-to-peer trust, while simultaneously recognizing several technological limitations and scalability difficulties.

A small group of hybrid or evolving structures is found in the remaining studies. These papers propose combining centralized registries with decentralized validation, layering AI based decision tools on top of blockchain systems, or enabling conditional access protocols based on stakeholder roles. While these models are not dominant, they reflect the field's experimentation with architectures that blend control with adaptability.

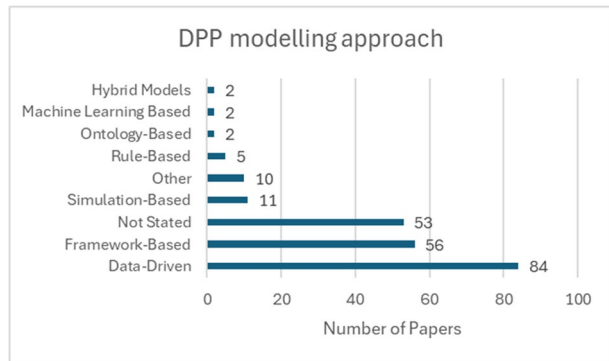
Overall, the architectural discourse indicates that DPPs are not a one-size-fits-all solution. Instead, system structure is closely tied to the intended application, regulatory landscape, and degree of stakeholder coordination. The coexistence of centralized, decentralized, and hybrid approaches suggests that architectural flexibility will remain a defining feature of DPP development going forward.

Modeling Approaches

Modeling approaches in DPP research vary significantly in purpose, ranging from system-level architecture design to analytical methods used for prediction, simulation, or decision support. While many papers do not specify a formal modeling method, those that do tend to fall into three primary categories: data driven, simulation based, and hybrid approaches (Fig. 10).

A total of 49 papers were coded as applying or proposing data driven models. These models frequently incorporate blockchain, IoT, or cloud based data systems to gather, structure, and analyze information across the product lifecycle. Data driven approaches are often used to create traceability records, monitor carbon emissions, or manage lifecycle events

Fig. 10 DPP modelling approach used in the analyzed papers



such as repair, reuse, and recycling. Machine learning is mentioned in some cases to support classification, anomaly detection, or lifecycle prediction.

Simulation based models appear in 16 papers, most commonly within case studies or technical feasibility assessments. These studies use simulations to test DPP behavior under specific scenarios, such as product returns, component disassembly, or recycling flows. These models are frequently used in circular economy studies and lifecycle analysis, where controlled virtual environments enable system behavior to be observed under varying assumptions.

Hybrid models are less common, with 9 papers combining data driven analytics with simulation or optimization frameworks. These studies aim to integrate real-time data into decision support tools, allowing for dynamic responses to lifecycle events. Such models are especially relevant in logistics heavy applications or where predictive maintenance is being explored.

A large number of papers, more than half of the total, do not specify any modeling approach. This absence is particularly evident in conceptual and review papers, where the emphasis is on policy alignment, system proposals, or architectural frameworks rather than operational modeling.

DPP research includes contributions from engineering, computer science, supply chain management, and sustainability, as shown by the wide range of modeling methods used. However, the heterogeneity of modeling approaches indicates insufficient standardization. This shows the importance of clearly classifying and testing suggested models.

Technological Infrastructure

The technological backbone of DPP systems is a focal point across much of the literature, with recurring references to blockchain, IoT, cloud platforms, and digital identifiers such as RFID or QR codes. These technologies are not only proposed as enablers but often define the feasibility and scope of DPP implementation.

Blockchain is the most frequently cited technology, appearing in 89 papers. It is typically proposed as a solution for decentralization, data immutability, and multi actor trust. Use cases range from tracking recycled content in plastics to verifying origin in textiles and managing battery compliance. While many of these papers position blockchain as a default choice for distributed data handling, several also note concerns related to energy intensity, scalability, and technical complexity.

RFID and QR codes are discussed in 62 papers as mechanisms for linking physical products to their digital counterparts. These identifiers serve as gateways between products and DPP platforms, enabling real time access to data at different lifecycle stages. RFID is particularly common in fashion and packaging sectors, while QR codes are proposed for consumer facing applications where mobile interaction is key.

IoT infrastructure appears in 41 papers, primarily in connection with data collection, real time monitoring, and sensor integration. IoT is often embedded into DPP systems in high value or high risk sectors like electronics, energy, and aerospace, where condition monitoring and predictive analytics are important for asset management and decision making Fig. 11.

Cloud based platforms are discussed in 31 papers, typically as hosting environments for DPP data and applications. These platforms are valued for their scalability, remote accessibility, and support for user specific data views. Several studies highlight cloud solutions as more accessible alternatives to blockchain, especially for small and medium enterprises.

Artificial intelligence and machine learning are mentioned in 29 papers, usually in support of predictive maintenance, anomaly detection, or lifecycle optimization. These capabilities are still emerging but indicate a growing trend toward smart DPPs capable of adaptive and automated decision support.

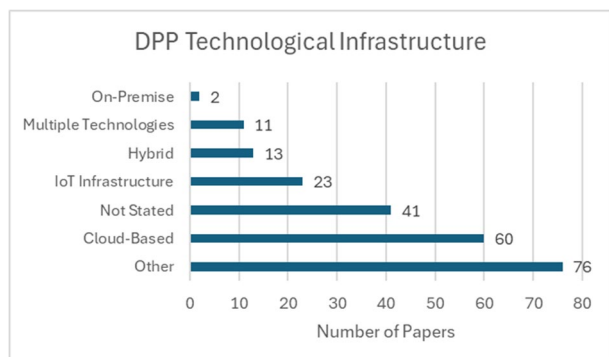
Other technologies appear less frequently but suggest expanding interest in diverse infrastructures. These include digital twins, cyber physical systems, and knowledge graphs, each mentioned in a small number of exploratory or technical studies.

The distribution of enabling technologies indicates a convergence of digital innovation domains around DPP development. However, it also reflects fragmentation, with many studies proposing siloed or sector specific architectures. This underscores the need for integrative approaches that align technical infrastructure with regulatory goals and lifecycle data demands.

Standards and Regulatory Alignment

Alignment with existing or emerging standards and regulations is a central theme across the DPP literature. A total of 66 papers explicitly discuss European Union policies, with many referencing the EU Battery Regulation, Ecodesign for Sustainable Products Regulation (ESPR), and the broader European Green Deal. These papers often treat regulatory

Fig. 11 DPP technological Infrastructure



requirements not just as drivers but as design constraints, shaping how DPPs must handle data structure, accessibility, and transparency across product lifecycles.

ISO and IEC standards are referenced in 21 papers, typically in relation to data formats, lifecycle assessment methods, or digital traceability frameworks. ISO standards related to environmental management (such as ISO 14040) and product information (such as ISO 10303 for STEP files) are often cited as technical foundations or complementary tools for DPP deployment. These references suggest a growing effort to ground DPP systems within globally recognized standards, even when regulatory environments vary Fig. 12.

Sector specific regulations are discussed in 18 papers, particularly in the context of automotive, construction, and electronics. Examples include vehicle end-of-life directives, building certification requirements, and RoHS/WEEE directives for electronics. These papers emphasize the need for DPPs to integrate seamlessly into compliance regimes, serving both internal decision makers and external auditors.

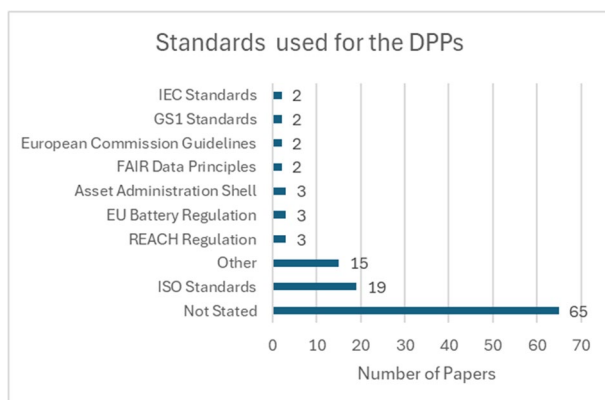
Digital standards and interoperability protocols are addressed in 15 papers, most of which call for harmonized data models and common vocabularies. These papers highlight the absence of shared standards as a major challenge and often propose ontologies, taxonomies, or open data frameworks to support cross sectoral and cross border operability.

Despite increasing alignment with formal regulations, 11 papers note a lack of legal clarity or fragmentation in the regulatory landscape. These studies point to inconsistent enforcement, slow standardization progress, and jurisdictional differences that complicate multinational adoption. Some also emphasize the risk of parallel or incompatible DPP systems emerging without clear governance.

The review suggests that while regulatory alignment is a strong driver of DPP development, the absence of unified standards across domains continues to hinder implementation. Efforts to link DPPs to international standards and EU level policy are advancing, but widespread harmonization remains an unresolved challenge.

An analysis of the modeling languages used across the reviewed literature reveals a significant gap in explicit data modeling practices within the DPP ecosystem. Of the 225 reviewed papers, more than half (125) do not specify any modeling language, indicating a lack of formalization or standardization in data representation across studies. This absence may suggest that many current works on DPPs remain conceptual or architectural in nature, without deep implementation or technical specification layers. The second largest category, labeled as “Other” (60 papers), also reflects fragmentation, implying that diverse and often

Fig. 12 Standards used for the DPP implementation



proprietary or context-specific modeling approaches are being explored without convergence toward recognized standards Fig. 13.

Among the studies that did state a modeling approach, lightweight data formats such as JSON (13 papers), RDF (10), and XML (9) are the most common. These languages are widely used in web technologies and are suitable for structuring flexible, interoperable data exchanges, especially in decentralized or cloud-based DPP platforms. The relatively low use of RDF, despite its suitability for semantic data and ontology-based traceability, may indicate a missed opportunity for enabling rich, machine-interpretable product metadata. The limited appearance of Asset Administration Shell (AAS) (3 papers), BIM Systems and IFC (2 each), and UML (1 paper) points to an underutilization of more formalized, industrial modeling frameworks. This is particularly notable given the relevance of AAS in Industrie 4.0 and of BIM/IFC in the construction sector.

These findings highlight the need for greater standardization and adoption of structured modeling languages to ensure DPP interoperability, especially in cross-sector and cross-border implementations. The current diversity of data structures poses a barrier to integration and automated reasoning, both of which are essential for scalable and robust DPP systems. Future research and policy initiatives should encourage the development or adoption of common data models preferably grounded in open standards like AAS, RDF/OWL, or industry-specific formats to foster harmonization across the rapidly evolving DPP landscape.

The “Other” category, which accounts for 60 papers, captures a diverse set of niche and sector-specific modeling approaches that fall outside mainstream standards. These include OPC-UA (Open Platform Communications – Unified Architecture), a machine-to-machine communication protocol widely used in Industrie 4.0 environments to support interoperability across heterogeneous manufacturing systems, and EPCIS (Electronic Product Code Information Services), a GS1 standard for capturing and sharing supply chain event data that enables visibility into product movement and status. In the construction domain, some papers reference IFC-SPF (Industry Foundation Classes – STEP Physical File format) as an alternative to standard IFC models, primarily for structured data exchange in building projects. A number of studies propose semantic data models without anchoring them to formal ontologies, seeking to represent product-related knowledge in a lightweight but

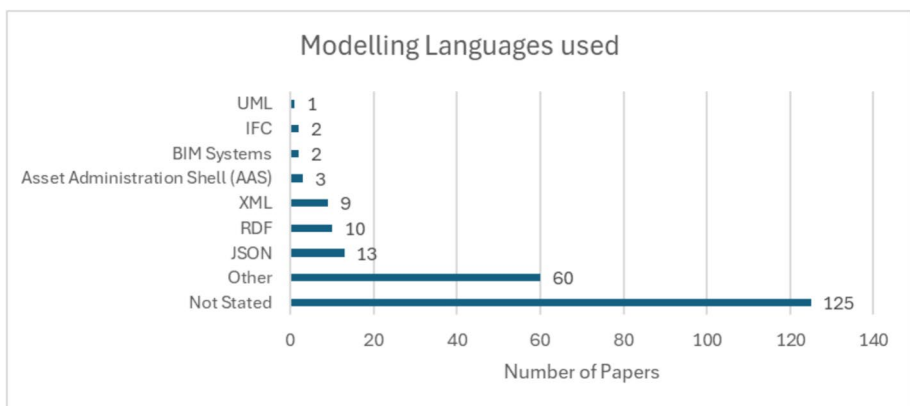


Fig. 13 Modelling languages used

less standardized form. Others develop custom frameworks for integrating carbon footprint metrics or broader sustainability indicators into product data structures, reflecting the regulatory pressure for environmental transparency. Experimental contributions introduce invariant learning frameworks, using AI-driven representations to support adaptive and automated structuring of DPP data. In addition, some papers adopt proprietary formats and GAIA-X semantic connectors, demonstrating how federated data-sharing environments can preserve sovereignty while enabling controlled interoperability across companies. Finally, several studies pursue framework-based categorization, mapping predefined attributes such as product characteristics, utilization, value chain processes, and sustainability metrics into structured information models. Collectively, these heterogeneous approaches highlight the creativity and sectoral tailoring present in the DPP ecosystem, but they also underscore the lack of convergence toward widely recognized standards, an issue that risks undermining interoperability and scalability across industries.

Scalability of DPP Implementations

Scalability of DPP implementations remains a central concern across the reviewed literature, though the way it is reported varies considerably (Fig. 14). The majority of papers (93 studies) fall into the *medium scalability* category, meaning that most DPP efforts have progressed beyond conceptual frameworks and laboratory studies but remain largely confined to pilots, sector-specific projects, or regionally limited implementations. These initiatives demonstrate partial stakeholder engagement and infrastructure readiness, yet full-scale operationalization across industries remains incomplete.

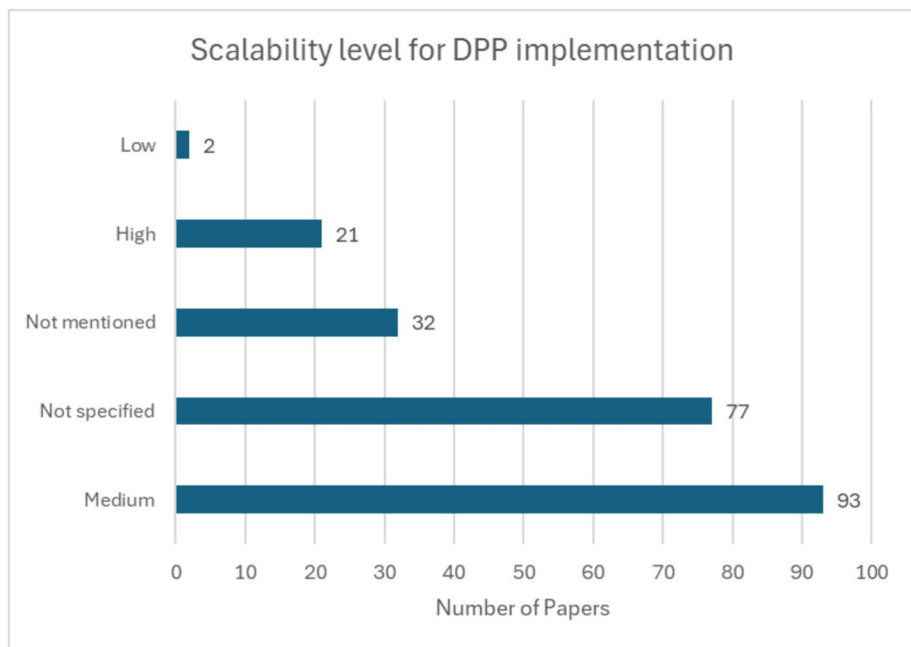


Fig. 14 Scalability levels

By contrast, only **21 studies** were classified as *high scalability*, typically linked to frameworks and infrastructures designed for multi-sector adoption or regulatory alignment. Examples include DPPs in the automotive battery sector, where regulatory drivers such as the EU Battery Regulation require deployment at a pan-European scale. A very small number of contributions (**2 papers**) were coded as *low scalability*, usually describing conceptual or early laboratory prototypes without evidence of deployment in practice.

In addition, **77 papers** fell under the category *not specified*, indicating that scalability was not discussed in sufficient detail to allow classification. Another **32 papers** were grouped as *not mentioned*, reflecting studies that made no reference to scalability at all. Together, these two groups highlight that while scalability is frequently recognized as an important issue, it is still often underexplored in the literature.

For clarity, we standardized these five categories as follows:

- **Low scalability:** Conceptual designs or lab prototypes without real-world testing.
- **Medium scalability:** Pilots, case studies, or sector-specific initiatives with limited adoption.
- **High scalability:** Frameworks or infrastructures designed for cross-sector or regulatory-driven deployment.
- **Not specified:** Scalability referenced but without sufficient detail for classification.
- **Not mentioned:** No reference to scalability at all.

This classification illustrates that the bulk of the literature situates DPP scalability at a transitional stage, beyond theory, but not yet reaching large-scale implementation [38]. It also underscores the need for more systematic evaluations of scalability as DPP initiatives move from pilot projects to regulatory enforcement and cross-sector adoption.

Beyond the clear classifications of low, medium, and high scalability, a substantial share of the literature did not provide enough information to allow precise categorization. In 77 papers, scalability was not specified: the issue was acknowledged but described only in vague terms such as “identified,” “requires,” or “designed,” without systematic analysis of how scalability could be achieved. These contributions signal an awareness of the importance of scalability, but they stop short of operationalizing it into concrete strategies or metrics. Another 32 papers fell into the not mentioned category, making no reference to scalability at all. This omission is noteworthy, as the ability to scale is essential for moving DPPs from conceptual discussion to real-world implementation, and its absence raises concerns about the maturity and practical relevance of such proposals.

At the other end of the spectrum, 21 studies explicitly demonstrated high scalability, most of them linked to regulated domains such as automotive and electronics. These cases often leverage decentralized or cloud-based infrastructures to enable cross-border data exchange and interoperability, reflecting both technological readiness and strong regulatory drivers (e.g., the EU Battery Regulation). By contrast, only 2 papers were classified as low scalability, typically describing conceptual or laboratory prototypes with limited applicability beyond controlled environments. These studies frequently point to fragmented data sources, weak stakeholder coordination, or a lack of legal harmonization as barriers that prevent wider adoption.

Overall, the way scalability is discussed across the literature, from vague acknowledgments to detailed architectural claims, underscores the absence of a unified assessment

framework. Future work would benefit from the establishment of standardized scalability indicators that address technical scalability (e.g., system performance, data handling capacity), organizational scalability (e.g., governance structures, stakeholder integration, SME participation), and regulatory scalability (e.g., compatibility with legal frameworks and cross-jurisdictional requirements). Developing such indicators, alongside systematic reporting of pilot outcomes and lessons learned, will be critical for enabling the transition from today's medium-scalability pilots to tomorrow's high-scalability, cross-sectoral DPP implementations capable of delivering sustainable and circular outcomes at scale.

Stakeholders and Governance Models

The effective implementation of Digital Product Passports (DPPs) depends not only on technological design but also on the active involvement of a wide range of stakeholders across the value chain. Because DPPs are intended to serve as integrative data infrastructures, their success hinges on the participation of both public and private actors, each contributing distinct roles, responsibilities, and incentives. Mapping stakeholder involvement across the literature provides insights into how DPP ecosystems are currently envisioned, which groups are prioritized, and where critical gaps in engagement remain. The analysis reveals clear asymmetries, with manufacturers and regulators dominating most discussions, while other actors such as consumers, SMEs, and service providers receive comparatively limited attention.

Stakeholder involvement is a critical dimension of DPP development, with most studies acknowledging the complex interplay between public and private actors (Fig. 15). The analysis reveals that 180 papers explicitly reference manufacturers as central stakeholders. Manufacturers are consistently framed as both primary data providers and key beneficiaries of DPP systems, particularly in contexts where traceability, compliance, and lifecycle optimization create operational and reputational value. Several studies stress the importance of establishing stronger incentives and clearer frameworks to encourage participation from manufacturers, especially among SMEs (11 papers), who face resource constraints and lack of technical expertise.

Regulators are identified in 44 papers and policymakers in 71 papers, often portrayed as initiators and enablers of DPP ecosystems. These actors are tasked with defining minimum data requirements, establishing interoperability standards, and supervising enforcement. In EU-focused studies, regulators and policymakers frequently appear as both rule setters and coordinators, particularly in centralized or semi-centralized architectures. Government bodies more broadly are mentioned in 51 papers, reflecting their influence in funding, oversight, and policy implementation.

Recyclers emerge in 70 papers, highlighting their role as downstream users of DPP data. Studies emphasize the critical need for standardized and accessible information on material composition, disassembly instructions, and hazardous content to support efficient recycling and material recovery. DPP integration with reverse logistics and take-back schemes is frequently highlighted as a way to close material loops.

Consumers appear in 40 papers, though their role is more fragmented compared to other stakeholders. Some studies propose DPPs as tools for consumer empowerment, enabling informed purchasing decisions and sustainability comparisons. Others warn about risks

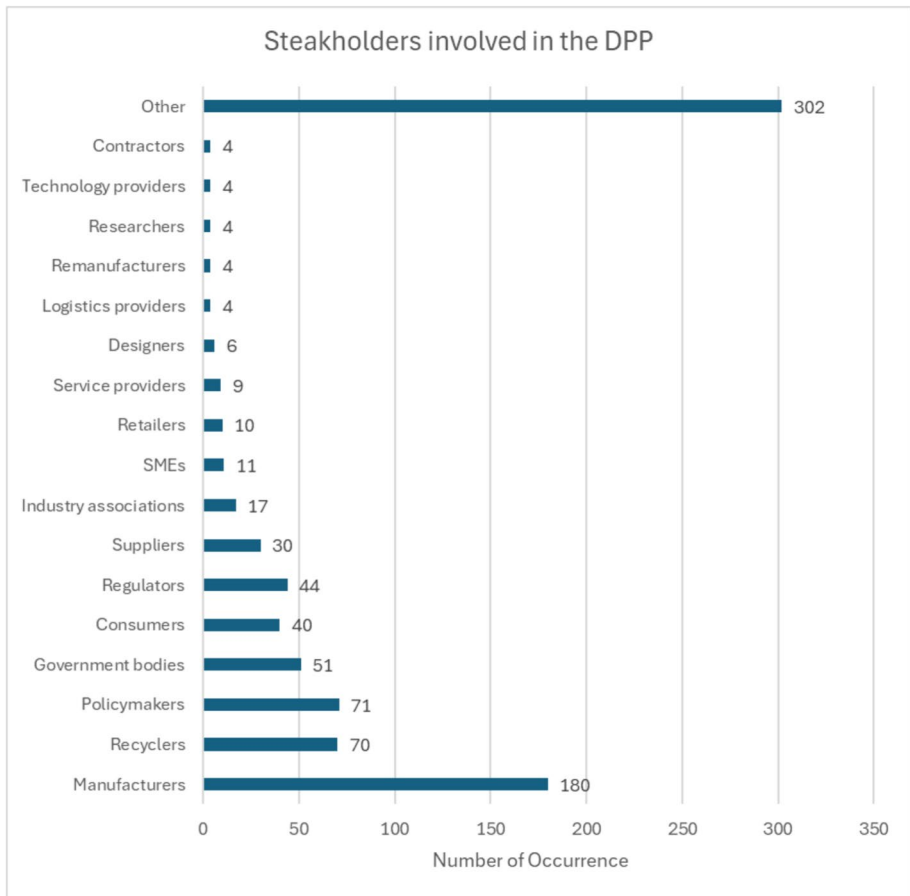


Fig. 15 Stakeholder types, involved in the DPPs

of data overload and limited usability, pointing to the need for human-centered design to ensure that consumer-facing features are intuitive, accessible, and actionable.

Other stakeholders are addressed less frequently. Suppliers are mentioned in 30 papers, generally in relation to upstream data provision and raw material traceability. Industry associations appear in 17 papers, often acting as intermediaries to facilitate sector-wide adoption. Retailers (10 papers) and service providers (9 papers) are recognized for their role in bridging product-level data with end-user services. A small set of papers highlights the contributions of designers (6), logistics providers (4), remanufacturers (4), researchers (4), technology providers (4), and contractors (4), though their roles remain underexplored in the general frameworks.

Governance models proposed across the literature vary in structure and complexity. Many studies advocate for multi-stakeholder governance frameworks that balance responsibilities between regulators, industry consortia, and third-party validators. Such models aim to ensure both compliance and innovation while fostering trust among stakeholders. Persistent challenges include data ownership, accountability, and clarity of roles. Several studies

stress the importance of co-design processes and iterative pilot projects to align interests and responsibilities before full-scale rollout.

Finally, the “Other” category (302 occurrences) consolidates a wide variety of actors that appear only sporadically in the literature. This includes NGOs, certification bodies, auditing organizations, standardization committees, financial institutions, and consumer advocacy groups. While each is mentioned infrequently, their inclusion reflects recognition that DPPs extend beyond traditional supply chain actors to a broader ecosystem of governance, finance, and societal stakeholders. Grouping these together ensures readability of results, though their diversity underscores the need for future research to better capture and integrate these complementary roles.

Advantages and Opportunities

The literature highlights a range of potential benefits associated with the implementation of DPPs. The most frequently cited advantage is improved lifecycle transparency, discussed in 88 papers.

The support for models of circular economies is the second most prevalent advantage, which was documented in 69 different articles. DPPs are positioned as major enablers of reuse, remanufacturing, and recycling as a result of these contributions, particularly because they provide information on product composition, repairability, and disassembly. Studies in this category frequently link DPPs to reverse logistics, eco-design, and material recovery strategies.

Regulatory compliance is mentioned in 56 papers, where DPPs are described as mechanisms to streamline reporting and ensure alignment with evolving legislation. In these papers, DPPs are not just tools for internal management but also function as verifiable interfaces between businesses and authorities, facilitating transparency and audit readiness.

Enhanced stakeholder collaboration appears in 39 papers as a strategic opportunity. These studies argue that by providing a shared data infrastructure, DPPs can bridge communication gaps among supply chain actors, including OEMs, recyclers, and regulators. This is especially relevant in complex global networks where data asymmetry hinders cooperation.

Innovation potential is noted in 27 papers, with authors suggesting that DPPs can serve as platforms for digital innovation in areas such as predictive maintenance, servitization, and value chain analytics. Some studies argue that DPPs could eventually evolve into smart product systems, interacting dynamically with other digital tools in manufacturing and logistics.

A smaller subset of papers also mention market differentiation, consumer trust, and brand reputation as additional benefits, particularly in consumer-facing sectors like fashion, electronics, and food. These advantages are typically tied to transparency efforts and the ability to communicate product sustainability in verifiable ways.

While these opportunities are widely recognized, most papers also acknowledge that realizing them depends on addressing substantial challenges in technology, governance, and standardization areas detailed in the following section.

Challenges and Barriers

Despite the promise of DPPs, the reviewed literature consistently identifies several barriers to their development and implementation. The most frequently cited challenge is data-related complexity, discussed in 76 papers. These studies highlight data silos throughout supply chains, fragmented data sources, a lack of uniform formats, and insufficient granularity. Data ownership and confidentiality issues are also common, particularly in industries with close competition where revealing information could be strategically damaging.

There are 59 publications that discuss legal fragmentation and regulatory uncertainty. Many of these studies draw attention to legal discrepancies, gaps in enforcement systems, and uncertainty around compliance standards. One factor contributing to and creating uncertainty is the dynamic character of EU laws like the ESPR and Battery Regulation. Several papers argue that without clear, harmonized mandates, DPP implementation may diverge across industries and regions, reducing overall effectiveness.

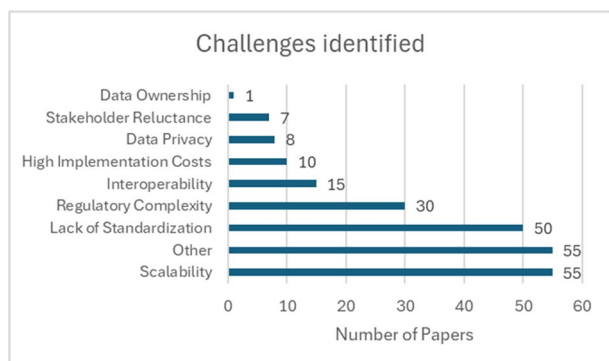
Technical limitations are identified in 47 papers, particularly regarding system scalability, cybersecurity, and infrastructure readiness. While blockchain and IoT are widely proposed as enabling technologies, many authors caution that their integration into industrial systems remains complex and cost-intensive. Interoperability with legacy systems and the need for flexible, upgradable platforms are recurring concerns (Fig. 16).

Economic burden and lack of incentives are discussed in 39 papers, with particular focus on small and medium-sized enterprises. These studies suggest that DPPs may introduce new costs related to data collection, platform subscription, or compliance auditing, without immediate return on investment. Several papers propose public support mechanisms, subsidies, or shared infrastructure to offset these burdens.

Stakeholder resistance and coordination difficulties appear in 28 papers. These challenges include unclear roles and responsibilities, conflicting interests between actors, and low trust among participants. Some studies emphasize the need for participatory design processes and pilot programs to build consensus and demonstrate early value.

A smaller group of 16 papers discuss consumer-side limitations, such as low digital literacy, privacy concerns, and uncertainty about how DPP information will be presented or interpreted. While consumers are often framed as potential beneficiaries of DPP transparency, many studies point out that current proposals lack clear communication strategies or user engagement mechanisms.

Fig. 16 Challenges identified



Sustainability Impact

Sustainability has emerged as the primary driver for the development and adoption of Digital Product Passports (DPPs). By design, DPPs are expected to enhance circular economy practices, improve transparency, and enable resource efficiency across value chains. However, the extent to which these impacts are realized varies significantly across the literature. To provide clarity, this review classifies the sustainability impact of DPPs into three categories: **high, medium, and low impact**, based on the depth and scope of reported outcomes. This structured classification allows for a more nuanced understanding of how far current DPP research and practice have progressed toward delivering measurable sustainability benefits.

Definitions of Impact Levels

- **High impact:** Studies that demonstrate or project substantial sustainability outcomes, supported by evidence such as reduced carbon footprint, higher recycling efficiency, recovery of critical raw materials, or measurable improvements in lifecycle performance.
- **Medium impact:** Studies that report partial or sector-specific sustainability improvements, often limited to pilots, qualitative assessments, or sectoral case studies without broad-scale validation.
- **Low impact:** Studies that mention sustainability in principle but lack quantitative data, clear outcomes, or scalability, often remaining at the conceptual or early prototype stage.

High impact (188 papers; 83.6%) The majority of studies position DPPs as high-impact instruments for advancing sustainability, showing clear benefits across sectors. The most commonly reported outcomes include:

- **Waste reduction and material recovery:** In plastics, [15] demonstrate how traceability improves mechanical recycling and reduces mismanagement of post-consumer waste. In textiles, [5] highlight how blockchain-based DPP consortia can support transparency in fashion supply chains and promote sustainable practices, while [39] discuss early German DPP applications that include textiles and apparel, focusing on waste reduction and sustainable consumption. In construction, [30] and [40] emphasize selective demolition, reuse of structural components, and the integration of BIM/IFC models for efficient material recovery.
- **Recycling efficiency and resource recovery:** The battery and PV sectors are heavily represented. [41] and [36] show how DPP-enabled tracking of lithium, cobalt, and rare earth elements supports the EU Battery Regulation by improving recycling efficiency. [42] extend this logic to PV modules, arguing that DPPs enhance the reuse of silicon and critical raw materials in solar energy supply chains.
- **Carbon footprint reduction and emissions transparency:** Multiple studies emphasize lifecycle data integration for climate action. [43] demonstrate reductions in CO₂ emissions through DPP-driven optimization in remanufacturing. Jansen et al. [23] show how embedding emissions data into product passports enables compliance with Scope 3 reporting requirements, which are increasingly critical for global supply chains.
- **Circular business models and eco-design strategies:** Several works highlight how DPPs

underpin new value propositions. Psarommatis et al. [1] quantify material recovery and defect prevention in remanufacturing by aligning DPP data with Zero Defect Manufacturing strategies. Wiesner et al., [12] propose a reference architecture that improves energy and resource efficiency in manufacturing, showing how lifecycle data supports eco-design. In the fashion sector, [44] illustrate how DPPs enable consumer-facing transparency, facilitating more sustainable purchasing decisions.

Taken together, these examples illustrate the breadth of DPP applications across industries. From electronic components and plastics to construction and textiles, the evidence shows that DPPs are increasingly being operationalized as mechanisms for waste reduction, recycling efficiency, and emissions transparency. The convergence of sectoral pilots around common goals, resource efficiency, lifecycle extension, and regulatory compliance, strongly supports the classification of most DPP studies as high-impact sustainability enablers.

Medium impact (33 papers; 14.7%) A smaller but meaningful group of studies fall into this category. These contributions emphasize sustainability but provide only partial or sector-specific evidence, often within pilot projects, conceptual frameworks, or exploratory studies. For example, [45] propose a blockchain architecture for industrial applications that outlines potential sustainability benefits, but the work remains at an architectural level without quantifiable environmental validation. From a digital infrastructure perspective, [46] examine how material passports can extend the capability of component digital threads for lifecycle management, though their study is limited to proof-of-concept implementations. Similarly, [47] gathers expert perspectives on blockchain in the circular economy, identifying transparency and traceability gains, yet relying primarily on expert judgment rather than empirical performance data. Finally, [48] reviews the role of blockchain for trade in global value chains, emphasizing its sustainability potential but noting that concrete, measurable impacts remain scarce.

These medium-impact studies highlight that while the sustainability potential of DPPs and related digital infrastructures is widely acknowledged, their benefits are often described in conceptual terms or demonstrated only in small-scale trials. Advancing from these early-stage explorations toward systematic, evidence-based demonstrations of measurable sustainability outcomes, such as reduced emissions, higher recycling efficiency, or extended product lifecycles, remains a key challenge for the field.

Interestingly, no papers in the dataset were classified as “Low sustainability impact.” This indicates that, across the literature, DPPs are rarely perceived as having negligible or negative sustainability outcomes. Instead, even in early-stage or conceptual works, DPPs are generally framed as enablers of sustainability, though the strength of evidence varies significantly.

Overall, the analysis reveals a strong consensus that DPPs have the potential to deliver **high sustainability impact**: over 80% of the literature frames them as transformative tools for resource efficiency and circularity. However, most of these papers focus on conceptual potential or sectoral pilots, with relatively few providing standardized, quantitative evidence. Medium- and low-impact studies, though fewer, underline the importance of robust measurement frameworks. To bridge the gap between aspirational narratives and demonstrated outcomes, future work should emphasize **evidence-based sustainability indicators** such as carbon footprint reduction, recycling efficiency, and lifecycle extension. Establish-

ing such common metrics will be essential to validate DPPs as genuine enablers of sustainable and circular industry practices.

Summary of Findings and Future Directions

The results of this review reveal that DPP research is rapidly expanding but remains dominated by conceptual and policy-driven studies. While there is growing attention to application contexts, especially in manufacturing, automotive, construction, and textiles, most contributions continue to focus on frameworks, proposed structures, and regulatory alignment rather than large-scale deployment or impact assessment. Although the objectives of sustainability, traceability, and the circular economy are frequently mentioned, their operationalization frequently lacks empirical depth and methodological consistency.

Stakeholder integration and governance emerge as important intersecting themes. Although they are acknowledged as key players, manufacturers, regulators, and consumers sometimes lack clear roles and duties. According to the literature, adoption may stall or lead to implementations that are compartmentalized in the absence of inclusive governance models and clear legal frameworks.

Numerous advantages are acknowledged, including lifecycle transparency, improved collaboration, and support for circular practices but these are consistently accompanied by significant barriers. Data availability, regulatory fragmentation, infrastructure readiness, and cost burdens persist as challenges, particularly for smaller actors and decentralized supply chains.

Sustainability impact is a dominant narrative, with most papers referencing ecological benefits. Yet few studies incorporate measurable indicators, and even fewer conduct lifecycle or impact assessments to quantify those claims. As a result, future research will need to move beyond broad claims of environmental alignment to develop verifiable and scalable models.

Together, these findings suggest that the DPP field is entering a transitional phase, shifting from high-level design and advocacy toward practical implementation. Future work will benefit from deeper methodological rigor, expanded empirical validation, and cross-sectoral coordination. Pilot studies, stakeholder-centered design, and international standard-setting efforts will be essential for translating the promise of DPPs into tangible and equitable outcomes across global value chains.

DPP life cycle Framework

The analysis of the reviewed literature highlights that there is currently no uniform definition or implementation of the DPP life cycle. Approaches differ widely, ranging from minimalistic concepts to advanced frameworks. On the one hand, some studies describe DPPs as simple identifiers or QR codes that provide access to static product data, often limited to regulatory compliance or consumer-facing transparency. On the other hand, more sophisticated implementations attempt to integrate DPPs into broader digital infrastructures, enabling real-time data exchange across manufacturing, supply chains, and product use. However, most of these advanced approaches still primarily support forward manufacturing

processes, with limited attention to post-use stages such as remanufacturing, recycling, or secondary material markets.

A second limitation observed across the reviewed works is that while many papers claim DPPs enable information traceability, the steps after the usage phase remain underdeveloped. Only a minority of studies address how DPPs could support end-of-life decisions, disassembly, or recycling logistics. For example, in construction-related papers, DPPs are linked to BIM/IFC models for selective demolition, yet frameworks for re-integration of recovered materials are not fully detailed. Similarly, in the battery sector, while DPPs are recognized as enablers of regulatory compliance and recycling, the actual mechanics of how data is transferred from product use to recycling facilities are often left ambiguous.

The systematic literature review confirms this fragmentation, as a large share of the 225 analyzed papers do not specify a complete product life cycle model, focusing instead on isolated phases such as manufacturing or usage. Only a few explicitly incorporate remanufacturing or closed-loop recycling, and even fewer consider maintenance as a structured phase where DPP data is updated and reused. This suggests that while the role of DPPs as carriers of lifecycle information is broadly acknowledged, their dynamic use across multiple life stages remains underexplored and inconsistent in practice.

Finally, several proposals treat the DPP as a static artifact, a fixed dataset generated at the time of manufacturing, rather than as a dynamic and evolving passport capable of being updated, re-assembled, or disassembled as the product moves through different life stages. This restricts the DPP's value for circular economy strategies, where accurate and timely data on material condition, usage history, and repair events are essential for enabling remanufacturing, reuse, and recycling at scale.

To move beyond fragmented and static approaches identified in the literature, this work proposes a holistic framework for the DPP life cycle that not only spans the classical stages of product life but also embeds the principle of information inheritance (Fig. 17). In this

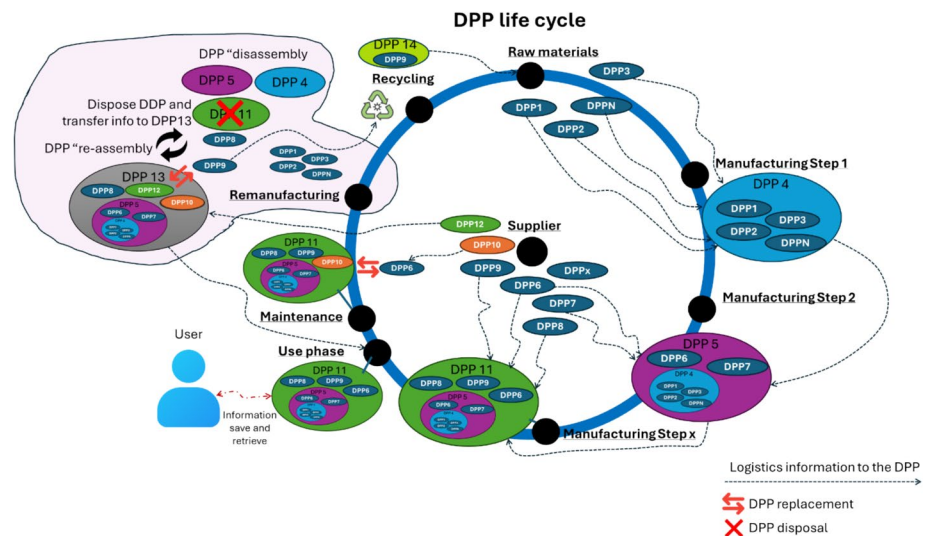


Fig. 17 DPP lifecycle Framework

framework, the DPP is not a static container, but a living, evolving digital artifact that grows and transforms as the product advances through its life stages.

At its inception, the DPP is created during the raw material extraction or sourcing stage, where it captures key attributes such as origin, certification, and sustainability indicators (e.g., carbon footprint, critical raw material content, social compliance). This first DPP establishes the foundation of traceability, providing the “seed” data upon which all subsequent product lifecycle information is built.

During the manufacturing stage, the DPP is enriched with process data, quality control results, and traceability information for components and assemblies. Here, information inheritance becomes crucial: each component arrives with its own DPP, and when assembled into a higher-level product, their data is not discarded but rather absorbed into the parent product’s DPP. This results in a hierarchical inheritance structure, where the DPP of a final product contains all relevant records from its subcomponents, creating a comprehensive digital twin of its history.

Once the product enters the usage phase, the DPP continues to expand. Performance data, usage conditions, maintenance logs, and even user-driven updates are recorded. Each event increases the “size” of the DPP, not necessarily in physical storage terms, but in informational depth. This cumulative growth reflects the reality that as a product’s life unfolds, its passport becomes richer, containing not only static design and production information but also dynamic, real-world operating history.

In the maintenance phase, DPPs serve as repositories for repair interventions, component replacements, and diagnostics. Importantly, when components are replaced, the new sub-component DPP is linked into the parent passport, while the outdated component’s DPP is “closed” but still archived for accountability. This ensures continuity and prevents information loss, creating an auditable chain of custody across component lifecycles.

The remanufacturing stage introduces the unique capability of DPP disassembly and re-assembly. When a product is disassembled, its DPP can be split into sub-passports for components or modules, each retaining its inherited information. When these parts are re-assembled into a remanufactured product, the component DPPs merge into a new product-level DPP, which consolidates all historical records into a unified document. This capability reflects real industrial practices, where reused parts carry forward their histories into new product configurations, ensuring transparency and reliability.

Finally, at recycling, the DPP plays a decisive role by providing data on material composition, disassembly instructions, hazardous substances, and potential reuse pathways. Even when the product is formally retired, its DPP does not vanish; instead, relevant information is transferred into new material passports that accompany recovered secondary raw materials. This ensures that knowledge generated during a product’s life is not lost but inherited by the next material cycle, reinforcing circular economy objectives.

An important dimension of this framework is the flexibility of DPPs to cover both batch and discrete product cases. In industries such as textiles, chemicals, food, and plastics, a single DPP may represent an entire batch, recording material properties, environmental impact, and processing history at the aggregate level. By contrast, in automotive, aerospace, or electronics, each discrete product (e.g., a car, a battery, or a laptop) is assigned its own DPP, often inheriting information from multiple component-level DPPs. For a DPP system to be effective, it must be capable of handling both granular unit-level traceability and aggregated batch-level tracking, with the possibility to link the two where required (e.g., a battery DPP

derived from a batch of cathode materials). This dual capacity ensures scalability across different sectors and supports the overarching goal of achieving transparent, interoperable, and circular product lifecycles.

Thus, the framework conceptualizes the DPP as a cumulative and dynamic system, where each new process, event, or transformation enlarges the passport. At the end of its journey, a DPP is not only a record of one product's life but a composite of all prior states, subcomponents, and inherited histories. This cumulative growth provides unprecedented opportunities for transparency, lifecycle optimization, and sustainability assessment, making the DPP a cornerstone of future circular industrial ecosystems.

Critical Discussion

Maturity and Orientation of DPP Research

The results of this review reveal that DPP research is rapidly expanding but remains dominated by conceptual and policy-driven studies. While there is growing attention to application contexts, especially in manufacturing, automotive, construction, and textiles, most contributions continue to focus on frameworks, proposed structures, and regulatory alignment rather than large-scale deployment or impact assessment. Although the objectives of sustainability, traceability, and the circular economy are frequently mentioned, their operationalization frequently lacks empirical depth and methodological consistency.

Taken together, these findings suggest that the DPP field is entering a transitional phase, shifting from high-level design and advocacy toward practical implementation. Future work will benefit from deeper methodological rigor, expanded empirical validation, and cross-sectoral coordination. Pilot studies, stakeholder-centered design, and international standard-setting efforts will be essential for translating the promise of DPPs into tangible and equitable outcomes across global value chains. Building on this synthesis, a more critical discussion is needed to evaluate the systemic barriers that currently hinder large-scale implementation of DPPs.

Governance Models and Stakeholder Integration

Stakeholder integration and governance also emerge as important intersecting themes. Although acknowledged as key actors, manufacturers, regulators, and consumers sometimes lack clear roles and duties. Adoption risks being fragmented in the absence of inclusive governance models and clear legal frameworks.

Another critical gap lies in stakeholder integration. While manufacturers and regulators dominate the discourse (referenced in 81 and 69 papers respectively), downstream actors such as consumers (51), recyclers (32), logistics providers, remanufacturers, and reuse actors remain underexplored. The role of end-users and recyclers is particularly marginalized, despite being essential for closing the loop in circular systems.

It should be noted, however, that consumer participation in the circular economy and sustainability requires different strategies compared to other stakeholders, and thus articles focusing specifically on consumer behavior were not captured within our search scope. Dedicated consumer-oriented studies (e.g., [43]) highlight this dimension and suggest that

future reviews should incorporate consumer-focused keywords to fully capture this critical perspective. Furthermore, few studies propose concrete governance frameworks that allocate clear roles and accountability across stakeholder groups.

Data and Information Modeling Gaps

A primary challenge is the lack of standardized data modeling practices, with 125 papers not specifying any modeling language and a further 60 using “Other” or unclear formats. Only a small minority employed structured formats like JSON (13), RDF (10), or XML (9), with negligible adoption of domain-relevant models like UML (1) or IFC (2).

This lack of formalism and alignment with international standards reflects an immature ecosystem where interoperability across platforms, actors, and borders remains aspirational. Without commonly accepted ontologies or modeling schemas, data silos will persist, limiting DPP portability, auditability, and cross-sector integration.

One of the most pressing challenges identified in this review is the lack of a clear and uniform methodology for allocating information to the correct DPP. The literature reveals a wide spectrum of interpretations: at one extreme, some proposals reduce the DPP to a minimalistic identifier or QR code linked to limited, static product data; at the other extreme, more advanced studies envision multi-layered frameworks integrated into digital infrastructures, capturing information across multiple lifecycle phases.

Yet in both cases, the absence of standardized rules for what information belongs in each passport and how it should be structured creates the risk of fragmentation. Critical lifecycle data may be recorded in the wrong context, lost in transition, or inconsistently updated, making continuity across the product lifecycle impossible.

Technological Architectures and Infrastructure Choices

Numerous advantages are acknowledged, including lifecycle transparency, improved collaboration, and support for circular practices, but these are consistently accompanied by significant barriers. Data availability, regulatory fragmentation, infrastructure readiness, and cost burdens persist as challenges, particularly for smaller actors and decentralized supply chains. Scalability, a prerequisite for DPP viability at industrial scale, is cited in varied and often vague terms. Only 21 papers report high scalability, while most fall into a medium category (93), suggesting limited deployments, often in testbeds or pilot settings. Forty-one papers merely identify scalability as a concern without detailed analysis.

The persistent lack of scalable infrastructure is compounded by architectural choices. Blockchain, although widely referenced (nearly 90 papers), is increasingly questioned for its operational feasibility. As highlighted by the results, blockchain’s energy intensity, infrastructure costs, and latency constraints make it ill-suited for high-frequency, real-time traceability across global networks. This raises serious concerns about its long-term viability, especially in carbon-sensitive industries and low-margin sectors. A hybrid architectural model combining blockchain or distributed ledgers for validation layers with centralized or cloud-based systems for general data storage is therefore a more pragmatic, energy-efficient, and flexible approach that should be prioritized in future DPP developments.

An important imbalance in the current DPP literature is the overwhelming emphasis on blockchain compared to the emerging role of Data Spaces. While blockchain offers benefits of transparency, immutability, and trustless verification, it also presents challenges related to scalability, energy use, and legal uncertainty. Data Spaces, by contrast, align more closely with European digital policy frameworks and are increasingly tested in large-scale pilots such as Catena-X and GAIA-X.

Future DPP infrastructures are therefore likely to combine the trust and provenance capabilities of blockchain with the interoperability and sovereignty guarantees of Data Spaces, creating hybrid solutions capable of scaling across multiple sectors. This dual perspective remains underrepresented in the literature but will be decisive for moving DPPs from pilot projects to regulatory enforcement and mass adoption.

On the technological front, enabling tools such as RFID, QR codes, and IoT sensors are frequently mentioned, particularly in fashion, packaging, and manufacturing. However, integration with backend infrastructure remains inconsistent. The presence of IoT in 41 papers and AI/ML in 29 signals a growing but still emergent trend toward “smart DPPs” capable of dynamic lifecycle tracking and predictive analytics. Yet very few studies offer robust use cases that link sensor data directly to automated lifecycle management or closed-loop decision-making frameworks.

Scalability and Industrial Deployment Challenges

Despite significant attention to standardization and regulatory alignment, compliance alone does not guarantee interoperability. While 144 papers declare alignment with existing regulations, primarily EU-focused, the presence of 66 non-compliant or unaligned works underscores the challenge of international regulatory fragmentation. Many studies tailor DPP architectures around specific regional legislation without considering global applicability. This raises concerns about regulatory islands and parallel frameworks, particularly problematic for industries with complex, multi-jurisdictional supply chains such as automotive, electronics, and fashion. The limited frequency of studies explicitly addressing future research directions further suggests that the field remains largely reactive and policy-driven. Only ten papers focus on forward-looking research agendas, indicating a dependence on regulatory mandates rather than proactive innovation. This dependency poses risks should policy directions diverge across jurisdictions.

Sustainability Impact: From Aspirational Claims to Measurable Outcomes

Sustainability impact is a dominant narrative, with most papers referencing ecological benefits. While 206 papers classify their work as having high sustainability impact, there is a glaring mismatch between rhetorical ambition and methodological rigor. Most studies fail to incorporate verifiable impact metrics such as carbon reduction or material recovery rates, or to apply Life Cycle Assessment methodologies. The absence of standardized KPIs and longitudinal measurement frameworks limits the ability to evaluate whether DPPs truly deliver on their circularity and climate claims. Several studies implicitly equate traceability with sustainability, overlooking the fact that visibility alone does not lead to improved environmental outcomes without corresponding design, procurement, or recovery actions.

Implications for Research, Policy, and Industrial Practice

The potential of DPPs to transform product lifecycles, enable decarbonization, and drive resource efficiency is clear. However, unlocking this potential will require a deliberate shift from scattered conceptual models to harmonized, data-driven, and scalable infrastructures grounded in open standards and inclusive governance.

The proposed framework (Fig. 17) addresses these shortcomings by positioning the DPP as a dynamic and evolving digital artifact based on the principle of information inheritance. Data from subcomponents and earlier life stages is absorbed into parent product passports, ensuring continuity across manufacturing, remanufacturing, and recycling. Equally important is the capability for disassembly and re-assembly of DPPs, allowing component-level traceability to persist across reuse and remanufacturing cycles. Without this modularity, traceability breaks down and downstream actors cannot trust the information chain.

A holistic approach is therefore required, uniting researchers, policymakers, and industry stakeholders around shared principles, including lifecycle coverage, interoperability, inheritance, and supply-chain-aware modeling. Establishing a minimum functional baseline for DPPs would prevent divergent interpretations, ensure comparability, and align implementations with the core goals of transparency and circularity. In summary, this critical discussion highlights systemic gaps in standardization, scalability, stakeholder integration, and impact assessment that must be addressed for DPPs to evolve from regulatory instruments into trusted enablers of sustainable industrial transformation.

Limitations of the study

This review has several limitations that should be acknowledged. First, although the study analyzes a large corpus of 225 peer-reviewed papers, it is inherently constrained by the quality, scope, and reporting practices of the underlying literature. Many studies lack detailed technical descriptions, empirical validation, or standardized metrics, which limits the depth of comparative analysis. Second, the review focuses on peer-reviewed academic publications and excludes grey literature, industrial white papers, and policy documents, which may contain relevant insights into real-world DPP deployments. Third, the analysis is predominantly based on English-language sources, potentially underrepresenting regional initiatives reported in other languages. Finally, while the classification framework captures a wide range of dimensions, the categorization of attributes such as sustainability impact and scalability necessarily involves interpretative judgment, despite efforts to apply consistent criteria. These limitations should be considered when interpreting the findings.

Future research directions

The findings of this review point to several priority directions for future research. First, there is a strong need for empirical and longitudinal studies that evaluate DPP implementations in operational settings, moving beyond conceptual frameworks and pilot demonstrations. Second, future work should focus on the development and adoption of standardized data and

information models, including ontologies and interoperable schemas, to support cross-sector and cross-border DPP integration. Third, research should investigate scalable and energy-efficient architectures, particularly hybrid approaches that combine distributed validation mechanisms with centralized or federated data infrastructures such as Data Spaces. Fourth, greater attention is required on downstream lifecycle phases, including maintenance, remanufacturing, reuse, and recycling, where the value of DPPs remains underexplored. Finally, future reviews should explicitly incorporate consumer behavior, usability, and human–DPP interaction, as these aspects are critical for closing material loops but remain insufficiently represented in the current literature. While cross-industry pilot projects covering full product lifecycles are critically needed to evaluate operational performance and implementation barriers of DPPs, such initiatives typically require multi-year industrial collaboration and therefore lie beyond the scope of review-based studies, reinforcing their role as a priority for future research rather than an expectation of the present work.

Policy and standardization implications

From a policy perspective, the review highlights that regulatory alignment alone is insufficient to ensure effective and interoperable DPP implementations. Policymakers should complement regulatory mandates with clear guidance on minimum functional requirements, data modeling standards, and governance principles for DPP systems. International coordination is particularly critical to avoid regulatory fragmentation and the emergence of incompatible regional DPP frameworks. Furthermore, policy instruments should account for the capacity constraints of small and medium-sized enterprises by promoting shared infrastructures, incentives, and phased adoption strategies. For standardization bodies, the results underscore the urgency of advancing open, lifecycle-oriented standards that address information inheritance, modularity, and interoperability across sectors. Aligning DPP-related standards with existing frameworks in digital manufacturing, lifecycle assessment, and data spaces will be essential to ensure that DPPs evolve from compliance tools into trusted infrastructures supporting circular economy and sustainability objectives.

Conclusions

This comprehensive review has examined the current state, challenges, and future directions of Digital Product Passports (DPPs) as enablers of sustainable manufacturing and circular economy transitions. By analyzing 225 academic and industry papers, this study presents the most detailed synthesis to date, classifying existing literature across multiple dimensions including regulatory compliance, technological architecture, modeling languages, industry application, sustainability impact, and implementation scalability.

The findings reveal that while DPPs hold significant promise for improving traceability, enhancing lifecycle transparency, and fostering regulatory alignment, their development remains fragmented and unevenly distributed across sectors. Many contributions present narrative or conceptual frameworks rather than empirically validated implementations, with scalability often restricted to pilots or sector-specific contexts. Moreover, technological debates remain dominated by blockchain, despite persistent concerns about energy ineffi-

ciency, operational costs, and interoperability challenges. Alternatives such as hybrid architectures and emerging data space initiatives offer more practical avenues, but they are still underexplored.

A key contribution of this review is the identification of the lack of a uniform DPP lifecycle methodology. Current practices vary widely, with some approaches reducing the passport to a static QR code linked to limited product data, while others propose sophisticated multi-layered infrastructures. The newly proposed DPP lifecycle framework addresses this gap by conceptualizing the passport as a dynamic, evolving digital artifact that grows with the product lifecycle. Through information inheritance, earlier lifecycle data is carried forward; through assembly and disassembly capabilities, passports can merge or split to represent complex supply chain interactions and remanufacturing pathways. This ensures information continuity and avoids the loss or misallocation of critical data during product transitions.

The review further highlights critical deficits in modeling standardization and sustainability measurement. More than half of the studies analyzed lack explicit data modeling languages, leading to fragmentation and limiting interoperability across platforms. Likewise, sustainability impacts are frequently claimed but rarely quantified with consistent indicators, leaving circularity outcomes unverifiable. Regulatory frameworks, particularly in the European Union, are advancing rapidly, yet global harmonization remains absent, creating uncertainty for international supply chains. In addition, stakeholder governance, consumer engagement, and end-of-life actors remain insufficiently addressed despite being central to achieving meaningful impact.

To transition from fragmented pilots to fully operational ecosystems, future research and policy should prioritize five actions:

1. Establishing internationally recognized data and modeling standards that support inheritance, modularity, and interoperability.
2. Developing energy-efficient, scalable infrastructures, including data spaces and hybrid architectures beyond blockchain.
3. Integrating DPPs with dynamic lifecycle assessment tools to ensure measurable sustainability outcomes.
4. Building inclusive governance frameworks that engage all stakeholders, from manufacturers to recyclers and consumers.
5. Validating DPP implementations through empirical, cross-industry pilots that demonstrate lifecycle-wide coverage, including reuse, remanufacturing, and recycling.

Only by addressing these systemic enablers and embracing the full lifecycle perspective can DPPs fulfill their transformative potential, not as narrow compliance instruments, but as foundational tools for resilient, transparent, and circular production systems. The limitations, future research needs, and policy implications identified in this study provide a roadmap for advancing DPPs toward scalable and interoperable industrial deployment.

All data generated or analyzed during this study are included in this published article

Annex 1

List of papers in the Systematic literature review

(Abdelmeguid et al., 2024; Adamkiewicz et al., 2022; Adisorn et al., 2021; Akhras et al., 2023b, 2023a; Albano et al., 2024; Alonso et al., 2023; Alt et al., 2023; Alves et al., 2024; Andisheh, 2024; André and Nilsson, 2024; Apel et al., 2024; Aragón and Alberti, 2024; Arshad et al., 2024; Bär et al., 2023; Barata and Kayser, 2024, 2023; Barkhausen et al., 2023; Barwasser et al., 2024; Baumgartner et al., 2024; Bellini et al., 2024a, 2024b; Berger et al., 2023d, 2023e, 2023b, 2023a, 2022; Bergonzoni et al., 2023; Bhubalan et al., 2022; Böhmecke-Schwafert, 2024; Boje et al., 2023; Boller et al., 2024; Boos et al., 2024; Boukhatmi et al., 2023; Bour et al., 2023; Bradley and Corsini, 2023; Brander et al., 2024; Bułkowska et al., 2023; Bundgaard and Huulgaard, 2023; Caldarelli, 2024; Carlsson and Nevzorova, 2024; Cassee et al., 2024; Centobelli et al., 2022; Çetin et al., 2023; Charnley et al., 2024; Chaudhuri et al., 2024; da Silva et al., 2024; Dalhammar et al., 2024; Davies et al., 2024; De Bruyne et al., 2023; Dervishaj and Gudmundsson, 2024; Dimitriou et al., 2024; Dragomir and Dumitru, 2024; Duarte et al., 2024; Ducuing and Reich, 2023; Ehmsen et al., 2024, 2023; Ermini et al., 2024; Esguerra et al., 2024; Fareed et al., 2024; Fiorello et al., 2023; Franke et al., 2024; Franzè et al., 2023; Freitas et al., 2024; Gallina et al., 2024, 2023; Geist and Balle, 2024; Gianvincenzi et al., 2024; Giovanardi et al., 2023; Gleich et al., 2024; Gómez-Gil et al., 2023; Greiner et al., 2024; Gupta and Jangra, 2024; Haswell et al., 2024; Haupt et al., 2024; Hedberg and Šipka, 2021; Heeß et al., 2024; Hoang and Böckel, 2024; Hoffmann et al., 2023; Horn et al., 2023; Hörner Bussolo et al., 2024; Hsu et al., 2024; Hulea et al., 2024; Jæger and Myrøld, 2023; Jakubowicz and Yarahmadi, 2024b; Jansen et al., 2023a, n.d.; H. H. Jensen et al., 2023; Jensen et al., 2024; S. F. Jensen et al., 2023b; Juhlin et al., 2024; Jung et al., 2024; Kalaboukas et al., 2023; Kaltenecker et al., 2024; Kebede et al., 2024, 2023; Kessler et al., 2023; Kim et al., 2023; King et al., 2023a; Klein et al., 2024; Koppelaar et al., 2023a; Kurniawan et al., 2024; Kuzminykh et al., 2024; Langley et al., 2023; A. Liu et al., 2024; H. Liu et al., 2024b, 2024a; Mallalieu et al., 2024; Mallick et al., 2024; Mallik, 2023; Marchesi et al., 2022; Mèda et al., 2024, 2023; Medaglia et al., 2024; Mendoza and Pigosso, 2023; Menny et al., 2024; Miron and Hulea, 2024; Morganti et al., 2024; Möslinger et al., 2023; Neri et al., 2024; Nissinen et al., 2022; Nowacki et al., 2023; Nyffenegger et al., 2024; Olsthoorn et al., 2023; Ospital et al., 2023; Ott et al., 2024; Pachimuthu et al., 2024; Palm et al., 2024; Panza et al., 2023; Papacharalampopoulos et al., 2024; Papamichael et al., 2024, 2023; Paramatmuni and Cogswell, 2023; Patrucco et al., 2024; Pehlken et al., 2024; Peiseler et al., 2023; Petelin, 2024; Picatoste et al., 2024; Piétron et al., 2023; Pinyol Alberich et al., 2023; Plociennik et al., 2024, 2022a, 2022c; Pohlmann et al., 2024a; Pomp et al., 2023; Popowicz et al., 2024; Pourjafarian et al., 2023; Pristerà et al., 2024; Psarommatis et al., 2024a, 2024b; Psarommatis and May, 2024a; Puglia et al., 2024; Pulikottil et al., 2024; Quinteros-Condoretty et al., 2025; Rebelo Moreira, 2024; Redeker et al., 2024; Reich et al., 2024, 2023; Rejeb et al., 2024; Ribeiro da Silva et al., 2023; Riemens et al., 2023; Rimaz et al., n.d.; Rizos and Urban, 2024; Rosado Da Cruz et al., 2024; Rukanova et al., 2024, n.d.; Rumetshofer et al., 2024a; Rumetshofer and Fischer, 2023a; Sá et al., 2024; Saenz et al., 2024b, 2024a; Sakao et al., 2024; Schauman et al., 2023; Schöggel et al., 2024, 2023; Schuberth et al., 2024a; Siska, 2023; Sousa et al., 2024; Spieß et al., 2024; Steiner and Münch, 2024; Steinwender et al., 2024; Stratmann et al., 2023; Stridsland et al., 2023; Stucki et al., 2024; Szaller et al., 2023; Tarashkar et al., 2024; Tarroja et al., 2024; Tell, 2024; Thomassen et al., 2024; Timms and King, 2023; Tolentino-Zondervan and DiVito, 2024; Tolio et al., 2023; Tong et al., 2024; Traverso et al., 2024; Tsunezawa et al., 2024; van Capelleveen et al., 2023; van Engelenburg et al., 2022; Van Opstal and Smeets, 2023; Vellesalu et al., 2023; Vomberg et al., 2024; Voukkali et al., 2024; Voulgaridis et al., 2023; Wagner et al., 2024a; Weber et al., 2024; Werner et al., 2025a; Wiesner et al., 2024a; Wu et al., 2024; Xu et al., 2024; Zante et al., 2024; Zhang and Seuring, 2024a; Zorpas, 2024)

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References

1. Psarommatis, F., Azamfierei, V., Lindström, J.D., 2024a. Circular product design for allowing re-using and re-purposing of products and components: a conceptual framework for circularity. IFAC-Pap., 6th IFAC Workshop on Advanced Maintenance Engineering, Services and Technology AMEST 2024 58, 306–311. <https://doi.org/10.1016/j.ifacol.2024.08.138>
2. Zhang A, Seuring S (2024) Digital product passport for sustainable and circular supply chain management: a structured review of use cases. *Int J Logist Res Appl* 27:2513–2540. <https://doi.org/10.1080/13675567.2024.2374256>
3. Koppelaar RHEM, Pamidi S, Hajósi E, Herreras L, Leroy P, Jung H-Y, Concheso A, Daniel R, Francisco FB, Parrado C, Dell'Ambrogio S, Guggiari F, Leone D, Fontana A (2023) A digital product passport for critical raw materials reuse and recycling. *Sustainability* 15:1405. <https://doi.org/10.3390/su15021405>
4. Chaudhuri A, Wæhrens BV, Treiblmaier H, Jensen SF (2024) Impact pathways: digital product passport for embedding circularity in electronics supply chains. *Int J Oper Amp Prod Manag.* <https://doi.org/10.1108/IJOPM-01-2024-0012>
5. Greiner M, Seidenfad K, Langewisch C, Hofmann A, Lechner U (2024a) The Digital Product Passport: Enabling Interoperable Information Flows Through Blockchain Consortia for Sustainability. In: Phillipson F, Eichler G, Erfurth C, Fahrnerberger G (eds) *Innovations for Community Services*. Springer Nature Switzerland, Cham, pp 377–396. https://doi.org/10.1007/978-3-031-60433-1_21
6. Voulgaridis K, Lagkas T, Angelopoulos CM, Boulogeorgos A-AA, Argyriou V, Sarigiannidis P (2024) Digital product passports as enablers of digital circular economy: a framework based on technological perspective. *Telecommun Syst* 85:699–715. <https://doi.org/10.1007/s11235-024-01104-x>
7. Berger K, Baumgartner RJ, Weinzerl M, Bachler J, Preston K, Schöggel J-P (2023) Data requirements and availabilities for a digital battery passport – a value chain actor perspective. *Clean Prod Lett* 4:100032. <https://doi.org/10.1016/j.clpl.2023.100032>
8. Jakubowicz I, Yarahmadi N (2024) Review and assessment of existing and future techniques for traceability with particular focus on applicability to ABS plastics. *Polymers* 16:1343. <https://doi.org/10.3390/polym16101343>
9. Jansen, M., Gerstenberger, B., Bitter-Krahe, J., Berg, H., Sebestyén, J., Schneider, J., 2022. Current approaches to the digital product passport for a circular economy: An overview of projects and initiatives. Wupp. Pap., Wuppertal Papers.
10. Plociennik C, Pourjafarian M, Nazeri A, Windholz W, Knetsch S, Rickert J, Ciroth A, Precci Lopes A, Hagedorn T, Vogelgesang M, Benner W, Gassmann A, Bergweiler S, Ruskowski M, Schebek L, Weidenkaff A (2022) Towards a digital lifecycle passport for the circular economy. *Procedia CIRP, The 29th CIRP Conference on Life Cycle Engineering*, April 4 – 6, 2022, Leuven, Belgium 105:122–127. <https://doi.org/10.1016/j.procir.2022.02.021>

11. Hariyani D, Hariyani P, Mishra S, Kumar Sharma M (2024) Leveraging digital technologies for advancing circular economy practices and enhancing life cycle analysis: a systematic literature review. *Waste Manag Bull* 2:69–83. <https://doi.org/10.1016/j.wmb.2024.06.007>
12. Wiesner M, Moreira J, Guizzardi R, Scholz P (2024a) A Reference Architecture for Digital Product Passports at Batch Level to Support Manufacturing Supply Chains. In: Araújo J, de la Vara JL, Santos MY, Assar S (eds) *Research Challenges in Information Science*. Springer Nature Switzerland, Cham, pp 307–323. https://doi.org/10.1007/978-3-031-59465-6_19
13. Digital Product Passport Market Size | Industry Report, 2030 [WWW Document], 2024. URL <http://www.grandviewresearch.com/industry-analysis/digital-product-passport-market-report> (accessed 9.5.25).
14. Alves L, Sá M, Cruz EF, Alves T, Alves M, Oliveira J, Santos M, Rosado da Cruz AM (2024) A traceability platform for monitoring environmental and social sustainability in the textile and clothing value chain: towards a digital passport for textiles and clothing. *Sustainability* 16:82. <https://doi.org/10.3390/su16010082>
15. Rumetshofer T, Fischer J (2023) Information-based plastic material tracking for circular economy, a review. *Polymers* 15:1623. <https://doi.org/10.3390/polym15071623>
16. Psarommatis F, May G (2024) Digital product passport: a pathway to circularity and sustainability in modern manufacturing. *Sustainability* 16:396. <https://doi.org/10.3390/su16010396>
17. Walden J, Steinbrecher A, Marinkovic M (2021) Digital product passports as enabler of the circular economy. *Chem Ing Tech* 93:1717–1727. <https://doi.org/10.1002/cite.202100121>
18. Jensen SF, Kristensen JH, Adamsen S, Christensen A, Waehrens BV (2023) Digital product passports for a circular economy: Data needs for product life cycle decision-making. *Sustain Prod Consum* 37:242–255. <https://doi.org/10.1016/j.spc.2023.02.021>
19. Jensen HH, Sornn-Friese H, Jensen SF, Aurisano N (2023) The Implications of Circular Supply Chains and the EU Digital Product Passport in Maritime Decarbonization. In: Lind M, Lehmacher W, Ward R (eds) *Maritime Decarbonization : Practical Tools, Case Studies and Decarbonization Enablers*. Springer Nature Switzerland, Cham, pp 231–250. https://doi.org/10.1007/978-3-031-39936-7_18
20. Adisorn T, Tholen L, Götz T (2021) Towards a digital product passport fit for contributing to a circular economy. *Energies* 14:2289. <https://doi.org/10.3390/en14082289>
21. King MRN, Timms PD, Mountney S (2023) A proposed universal definition of a Digital Product Passport Ecosystem (DPPE): Worldviews, discrete capabilities, stakeholder requirements and concerns. *J Clean Prod* 384:135538. <https://doi.org/10.1016/j.jclepro.2022.135538>
22. Plociennik C, Nazeri A, Rimaz MH, Knetsch S, Lopes AdoCP, Hagedorn T, Baehr J, Vogelgesang M, Li C, Benner W, Kellerer B, Ionescu E, Ruskowski M, Weidenkaff A (2024) Connecting Producers and Recyclers: A Digital Product Passport Concept and Implementation Suitable for End-of-Life Management. *Procedia CIRP* 122:831–836. <https://doi.org/10.1016/j.procir.2024.02.026>
23. Jansen M, Meisen T, Plociennik C, Berg H, Pomp A, Windholz W (2023) Stop guessing in the dark: identified requirements for digital product passport systems. *Syst* 11:123. <https://doi.org/10.3390/systems11030123>
24. Berger K, Rusch M, Pohlmann A, Popowicz M, Geiger BC, Gursch H, Schöggel J-P, Baumgartner RJ (2023) Confidentiality-preserving data exchange to enable sustainable product management via digital product passports - a conceptualization. *Procedia CIRP* 116:354–359. <https://doi.org/10.1016/j.procir.2023.02.060>. **(30th CIRP Life Cycle Engineering Conference)**
25. Berger K, Baumgartner RJ, Weinzerl M, Bachler J, Schöggel J-P (2023) Factors of digital product passport adoption to enable circular information flows along the battery value chain. *Procedia CIRP* 116:528–533. <https://doi.org/10.1016/j.procir.2023.02.089>. **(30th CIRP Life Cycle Engineering Conference)**
26. Hoffmann F, Koch T, Weigold M, Metternich J (2023) A data-based business concept to support product creation in reducing greenhouse gas emissions. *Procedia CIRP* 120:864–869. <https://doi.org/10.1016/j.procir.2023.09.089>
27. Plociennik, C., Pourjafarian, M., Saleh, S., Hagedorn, T., Carmo Precci Lopes, A. do, Vogelgesang, M., Baehr, J., Kellerer, B., Jansen, M., Berg, H., Ruskowski, M., Schebek, L., Ciroth, A., 2022c. Requirements for a Digital Product Passport to Boost the Circular Economy. https://doi.org/10.18420/INF2022_127
28. Moher D, Liberati A, Tetzlaff J, Altman DG (2010) Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Int J Surg* 8:336–341. <https://doi.org/10.1016/j.ijsu.2010.02.007>
29. Veroniki AA, Tricco AC, Rangira D, McKenzie JE, Li T, Straus SE, Smith M, Catalá-López F, Wolfe D, Nincic V, Konstantinidis M, Franco J, Tovey D, Thirugnanasampanthar SS, Dourka J, Warren R, Wells G, Stevens A, Hutton B (2025) Updating the PRISMA reporting guideline for network meta-analysis: a scoping review. *J Clin Epidemiol* 188:111985. <https://doi.org/10.1016/j.jclinepi.2025.111985>

30. Stratmann, L., Hoeborn, G., Pahl, C., Schuh, G., 2023. Classification of product data for a Digital Product Passport in the manufacturing industry 448–458. <https://doi.org/10.15488/13463>
31. Méda, P., Munir, M., Calvetti, D., Sousa, H., 2023. Information needs in digital products passport – discussing data framework and reasonability. Presented at the 2023 European Conference on Computing in Construction and the 40th International CIB W78 Conference. <https://doi.org/10.35490/EC3.2023.244>
32. Kebede, R., Moscati, A., Tan, H., Johansson, P., 2023. Circular economy in the built environment: a framework for implementing digital product passports with knowledge graphs, in: Computing in Construction. Presented at the EC3 Conference 2023, European Council on Computing in Construction, pp. 0–0. <https://doi.org/10.35490/EC3.2023.245>
33. Jæger B, Myrøld S (2023) Textile Industry Circular Supply Chains and Digital Product Passports. Two Case Studies. In: Alfnes E, Romsdal A, Strandhagen JO, von Cieminski G, Romero D (eds) *Advances in Production Management Systems. Production Management Systems for Responsible Manufacturing, Service, and Logistics Futures*. Springer Nature Switzerland, Cham, pp 350–363. https://doi.org/10.1007/978-3-031-43688-8_25
34. Spiß M, Berninger S, Perau M, Janßen J, Boos W, Schröer T (2024) Bidirectional Integration of Digital Product Passports into Information Systems of Production Planning and Control. In: Camarinha-Matos LM, Ortiz A, Boucher X, Barthe-Delanoë A-M (eds) *Navigating Unpredictability: Collaborative Networks in Non-Linear Worlds*. Springer Nature Switzerland, Cham, pp 179–193. https://doi.org/10.1007/978-3-031-71743-7_12
35. Hoang K-M, Böckel A (2024) Cradle-to-cradle business model tool: innovating circular business models for startups. *J Clean Prod* 467:142949. <https://doi.org/10.1016/j.jclepro.2024.142949>
36. Saenz J, Felsch T, Walter C, König T, Poenicke O, Bayrhammer E, Vorbröcker M, Berndt D, Elkmann N, Arlinghaus J (2024a) Automated disassembly of e-waste, requirements on modeling of processes and product states. *Front Robot AI*. <https://doi.org/10.3389/frobt.2024.1303279>
37. Saenz J, Felsch T, Walter C, König T, Poenicke O, Bayrhammer E, Vorbröcker M, Berndt D, Elkmann N, Arlinghaus J (2024b) Automated disassembly of e-waste, requirements on modeling of processes and product states. *Front Robot AI*. <https://doi.org/10.3389/frobt.2024.1303279>
38. Milat AJ, King L, Bauman AE, Redman S (2013) The concept of scalability: increasing the scale and potential adoption of health promotion interventions into policy and practice. *Health Promot Int* 28:285–298. <https://doi.org/10.1093/heapro/dar097>
39. Alt S, Schorr V, Vocke A, Werth D (2023) Status and future of real-life application of the digital process passport in Germany. *Circ Econ Sustain* 3:2255–2266. <https://doi.org/10.1007/s43615-023-00275-0>
40. Rebelo Moreira JL (2024) The Role of Interoperability for Digital Twins. In: Sales TP, de Kinren S, Proper HA, Pufahl L, Karastoyanova D, van Sinderen M (eds) *Enterprise Design, Operations, and Computing. EDOC 2023 Workshops*. Springer Nature Switzerland, Cham, pp 139–157. https://doi.org/10.1007/978-3-031-54712-6_9
41. Kim J, Moon J, Do Y, Kim H, Jeong J (2023) Research on a SaaS (Software as a Service)-based digital product passport system model for the EV battery industry value chain. *WSEAS Trans Bus Econ* 20:2466–2475. <https://doi.org/10.37394/23207.2023.20.211>
42. Piétron D, Staab P, Hofmann F (2023) Digital circular ecosystems: a data governance approach. *GAIA - Ecological Perspectives for Science and Society* 32:40–46. <https://doi.org/10.14512/gaia.32.S1.7>
43. Szaller Á, Gallina V, Gal B, Gaal A, Fries C (2023) Quantitative benefits of the digital product passport and data sharing in remanufacturing. *Procedia CIRP* 120:928–933. <https://doi.org/10.1016/j.procir.2023.09.102>. **(56th CIRP International Conference on Manufacturing Systems 2023)**
44. Duarte N, Pereira C, Grzywińska-Rapca M, Kulli A, Goci E (2024) Consumers' knowledge and decisions on circularity: Albanian, Polish, and Portuguese perspectives. *Environ Dev Sustain*. <https://doi.org/10.1007/s10668-024-05077-4>
45. Marchesi L, Marchesi M, Tonelli R, Lunesu MI (2022) A blockchain architecture for industrial applications. *Blockchain Res Appl* 3:100088. <https://doi.org/10.1016/j.bcr.2022.100088>
46. Paramatmuni C, Cogswell D (2023) Extending the capability of component digital threads using material passports. *J Manuf Process* 87:245–259. <https://doi.org/10.1016/j.jmapro.2023.01.032>
47. Caldarelli G (2024) Expert perspectives on blockchain in the circular economy: a Delphi study with industry specialists. *J Clean Prod* 465:142781. <https://doi.org/10.1016/j.jclepro.2024.142781>
48. Böhmecke-Schwafert M (2024) The role of blockchain for trade in global value chains: a systematic literature review and guidance for future research. *Telecomm Policy* 48:102835. <https://doi.org/10.1016/j.telpol.2024.102835>
49. Abdelmeguid A, Afy-Shararah M, Salonitis K (2024) Towards circular fashion: management strategies promoting circular behaviour along the value chain. *Sustain Prod Consum* 48:143–156. <https://doi.org/10.1016/j.spc.2024.05.010>
50. Adamkiewicz J, Kochańska E, Adamkiewicz I, Łukasik RM (2022) Greenwashing and sustainable fashion industry. *Curr Opin Green Sustain Chem* 38:100710. <https://doi.org/10.1016/j.cogsc.2022.100710>

51. Akhras, M.H., Fischer, J., Freudenthaler, P.J., 2023a. Mechanical Recycling of Polyethylene Bottle Caps - Characterization of the Various Material States along the Process Value Chain. *Proc. ANTEC* 2023.
52. Akhras MH, Freudenthaler PJ, Straka K, Fischer J (2023) From bottle caps to frisbee, a case study on mechanical recycling of plastic waste towards a circular economy. *Polymers* 15:2685. <https://doi.org/10.3390/polym15122685>
53. Albano J, Rahn A, Bachmann J, Wende G (2024) Life cycle inventories for aviation: background data, shortcomings, and improvements. *Clean Eng Technol* 22:100813. <https://doi.org/10.1016/j.clet.2024.100813>
54. Alonso R, Olivadese R, Ibba A, Reforgiato Recupero D (2023) Towards the definition of a European digital building logbook: a survey. *Heliyon* 9:e19285. <https://doi.org/10.1016/j.heliyon.2023.e19285>
55. Andisheh K (2024) Development of steel circularity passport: literature review, research gaps, and program rules in New Zealand. *Waste Manag Bull* 2:11–27. <https://doi.org/10.1016/j.wmb.2024.06.003>
56. André H, Nilsson L (2024) Are second-hand shell jackets better than users think? A comparison of perceived, assessed and measured functionality throughout lifespans. *Resour Conserv Recycl* 204:107470. <https://doi.org/10.1016/j.resconrec.2024.107470>
57. Apel C, Kümmerer K, Sudheshwar A, Nowack B, Som C, Colin C, Walter L, Breukelaar J, Meeus M, Ildefonso B, Petrovykh D, Elyahmadi C, Huttunen-Saarivirta E, Dierckx A, Devic AC, Valsami-Jones E, Brennan M, Rocca C, Scheper J, Strömberg E, Soeteman-Hernández LG (2024) Safe-and-sustainable-by-design: state of the art approaches and lessons learned from value chain perspectives. *Curr Opin Green Sustain Chem* 45:100876. <https://doi.org/10.1016/j.cogsc.2023.100876>
58. Aragón A, Alberti MG (2024) Limitations of machine-interpretability of digital EPDs used for a BIM-based sustainability assessment of construction assets. *J Build Eng* 96:110418. <https://doi.org/10.1016/j.jobe.2024.110418>
59. Arshad MH, Saxena KK, Reynaerts D (2024) Hybrid laser-ECM processing of advanced materials in neutral electrolyte without aggressive reagents: machining performance, removal characteristics and sustainability aspects. *J Clean Prod* 467:143034. <https://doi.org/10.1016/j.jclepro.2024.143034>
60. Bär, M.A., Colombo, A.W., Torres, J.L., Fernandez, E., Rico, M., Caliusco, M.L., 2023. An Industry 4.0-Compliant Digital Product Passport Approach for Realising Dairy Product Traceability, in: *IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society*. Presented at the *IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society*, pp. 1–9. <https://doi.org/10.1109/IECON51785.2023.10312481>
61. Barata J, Kayser I (2024) How will the digital twin shape the future of industry 5.0? *Technovation* 134:103025. <https://doi.org/10.1016/j.technovation.2024.103025>
62. Barata J, Kayser I (2023) Industry 5.0 – Past, Present, and Near Future. *Procedia Comput Sci* 219:778–788. <https://doi.org/10.1016/j.procs.2023.01.351>. (CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANAGEMENT / HCist – International Conference on Health and Social Care Information Systems and Technologies 2022)
63. Barkhausen R, Fick K, Durand A, Rohde C (2023) Analysing policy change towards the circular economy at the example of EU battery legislation. *Renew Sustain Energy Rev* 186:113665. <https://doi.org/10.1016/j.rser.2023.113665>
64. Barwasser A, Schuseil F, Werner A, Jung M, Zimmermann N (2024) The Digital Product Passport: Scenario-based Recommendations for the Manufacturing Industry. *IFAC-Pap* 58:1186–1191. <https://doi.org/10.1016/j.ifacol.2024.09.097>. (18th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2024)
65. Baumgartner RJ, Berger K, Schögl J-P (2024) Digital Technologies for Sustainable Product Management in the Circular Economy. In: Lynn T, Rosati P, Kreps D, Conboy K (eds) *Digital Sustainability: Leveraging Digital Technology to Combat Climate Change*. Springer Nature Switzerland, Cham, pp 121–143. https://doi.org/10.1007/978-3-031-61749-2_7
66. Bellini A, Andersen B, Klungseth NJ, Tadayon A (2024) Achieving a circular economy through the effective reuse of construction products: a case study of a residential building. *J Clean Prod* 450:141753. <https://doi.org/10.1016/j.jclepro.2024.141753>
67. Bellini A, Tadayon A, Andersen B, Klungseth NJ (2024) The role of data when implementing circular strategies in the built environment: a literature review. *Clean Environ Syst* 13:100183. <https://doi.org/10.1016/j.cesys.2024.100183>
68. Berger K, Schögl J-P, Baumgartner RJ (2022) Digital battery passports to enable circular and sustainable value chains: conceptualization and use cases. *J Clean Prod* 353:131492. <https://doi.org/10.1016/j.jclepro.2022.131492>

69. Bergonzoni M, Melloni R, Botti L (2023) Analysis of sustainable concrete obtained from the by-products of an industrial process and recycled aggregates from construction and demolition waste. *Procedia Comput Sci* 217:41–51. <https://doi.org/10.1016/j.procs.2022.12.200>. (4th International Conference on Industry 4.0 and Smart Manufacturing)
70. Bhubalan K, Tamothran AM, Kee SH, Foong SY, Lam SS, Ganeson K, Vigneswari S, Amirul A-A, Ramakrishna S (2022) Leveraging blockchain concepts as watermarkers of plastics for sustainable waste management in progressing circular economy. *Environ Res* 213:113631. <https://doi.org/10.1016/j.envres.2022.113631>
71. Boje C, Hahn Menacho AJ, Marvuglia A, Benetto E, Kubicki S, Schaubroeck T, Navarrete Gutiérrez T (2023) A framework using BIM and digital twins in facilitating LCSA for buildings. *J Build Eng* 76:107232. <https://doi.org/10.1016/j.jobe.2023.107232>
72. Boller ML, Zurwehme A, Krupitzer C (2024) Qualitative assessment on the chances and limitations of food fraud prevention through distributed ledger technologies in the organic food supply chain. *Food Control* 158:110247. <https://doi.org/10.1016/j.foodcont.2023.110247>
73. Boos W, Lukas G, Trisjono J, Eberius T, Klisch L (2024) Conceptualization of a methodology for circular value creation in SMEs in one-off production. *Procedia CIRP* 122:229–234. <https://doi.org/10.1016/j.procir.2024.02.008>
74. Boukhatmi A, Nyffenegger R, Grösser SN (2023) Designing a digital platform to foster data-enhanced circular practices in the European solar industry. *J Clean Prod* 418:137992. <https://doi.org/10.1016/j.jclepro.2023.137992>
75. Bour A, Budde Christensen T, Hunka AD, Palmqvist A, Skjold E, Syberg K (2023) Implications of circular textile policies for the future regulation of hazardous substances in textiles in the European Union. *Sci Total Environ* 896:165153. <https://doi.org/10.1016/j.scitotenv.2023.165153>
76. Bradley CG, Corsini L (2023) A literature review and analytical framework of the sustainability of reusable packaging. *Sustain Prod Consum* 37:126–141. <https://doi.org/10.1016/j.spc.2023.02.009>
77. Brander SM, Senathirajah K, Fernandez MO, Weis JS, Kumar E, Jahnke A, Hartmann NB, Alava JJ, Farrelly T, Almroth BC, Groh KJ, Syberg K, Buerkert JS, Abeynayaka A, Booth AM, Cousin X, Herzke D, Monclús L, Morales-Caselles C, Bonisoli-Alquati A, Al-jabachi R, Wagner M (2024) The time for ambitious action is now: science-based recommendations for plastic chemicals to inform an effective global plastic treaty. *Sci Total Environ* 949:174881. <https://doi.org/10.1016/j.scitotenv.2024.174881>
78. Bułkowska K, Zielińska M, Bułkowski M (2023) Implementation of blockchain technology in waste management. *Energies* 16:7742. <https://doi.org/10.3390/en16237742>
79. Bundgaard AM, Huulgaard RD (2023) The role of standards in support of material efficiency requirements under the Ecodesign Directive. *J Clean Prod* 385:135599. <https://doi.org/10.1016/j.jclepro.2022.135599>
80. Carlsson R, Nevzorova T (2024) Managing Circular Electric Vehicle Battery Lifecycles Using Standards. In: Scholz SG, Howlett RJ, Setchi R (eds) *Sustainable Design and Manufacturing 2023*. Springer Nature, Singapore, pp 63–77. https://doi.org/10.1007/978-981-99-8159-5_6
81. Cassee FR, Bleeker EAJ, Durand C, Exner T, Falk A, Friedrichs S, Heunisch E, Himly M, Hofer S, Hofstätter N, Hristozov D, Nymark P, Pohl A, Soeteman-Hernández LG, Suarez-Merino B, Valsami-Jones E, Groenewold M (2024) Roadmap towards safe and sustainable advanced and innovative materials. (Outlook for 2024–2030). *Comput Struct Biotechnol J* 25:105–126. <https://doi.org/10.1016/j.csbj.2024.05.018>
82. Centobelli P, Abbate S, Nadeem SP, Garza-Reyes JA (2022) Slowing the fast fashion industry: an all-round perspective. *Curr Opin Green Sustain Chem* 38:100684. <https://doi.org/10.1016/j.cogsc.2022.100684>
83. Çetin S, Raghu D, Honic M, Straub A, Gruis V (2023) Data requirements and availabilities for material passports: a digitally enabled framework for improving the circularity of existing buildings. *Sustain Prod Consum* 40:422–437. <https://doi.org/10.1016/j.spc.2023.07.011>
84. Charnley F, Cherrington R, Mueller F, Jain A, Nelson C, Wendland S, Ventosa S (2024) Retaining product value in post-consumer textiles: how to scale a closed-loop system. *Resour Conserv Recycl* 205:107542. <https://doi.org/10.1016/j.resconrec.2024.107542>
85. da Silva SBG, Barros MV, Radicchi JÁZ, Puglieri FN, Piekarski CM (2024) Opportunities and challenges to increase circularity in the product's use phase. *Sustainable Futures* 8:100297. <https://doi.org/10.1016/j.sfr.2024.100297>
86. Dalhammar, C., Richter, J., Montenegro, P., 2024. Drivers and Barriers for “Circular” Consumer Electronics in the European Union, in: *2024 Electronics Goes Green 2024+ (EGG)*. Presented at the 2024 Electronics Goes Green 2024+ (EGG), pp. 1–10. <https://doi.org/10.23919/EGG62010.2024.10631178>
87. Davies J, Sharifi H, Lyons A, Forster R, Elsayed OKSM (2024) Non-fungible tokens: the missing ingredient for sustainable supply chains in the metaverse age? *Transportation Research Part E: Logistics and Transportation Review* 182:103412. <https://doi.org/10.1016/j.tre.2024.103412>

88. De Bruyne J, Dheu O, Ducuing C (2023) The European Commission's approach to extra-contractual liability and AI – an evaluation of the AI liability directive and the revised product liability directive. *Comput Law Secur Rev* 51:105894. <https://doi.org/10.1016/j.clsr.2023.105894>
89. Dervishaj A, Gudmundsson K (2024) From LCA to circular design: a comparative study of digital tools for the built environment. *Resour Conserv Recycl* 200:107291. <https://doi.org/10.1016/j.resconrec.2023.107291>
90. Dimitriou DJ, Sartzetaki MF, Karagkouni AG (2024) Chapter One - Airport Operators Performance Management in the Era of Business Intelligence. In: Dimitriou DJ, Sartzetaki MF, Karagkouni AG (eds) *Managing Airport Corporate Performance*. Elsevier, pp 1–27. <https://doi.org/10.1016/B978-0-443-29109-8.00001-3>
91. Dragomir VD, Dumitru M (2024) The state of the research on circular economy in the European Union: a bibliometric review. *Clean Waste Syst* 7:100127. <https://doi.org/10.1016/j.clwas.2023.100127>
92. Ducuing C, Reich RH (2023) Data governance: digital product passports as a case study. *Compet Regul Netw Ind* 24:3–23
93. Ehmsen S, Klar M, Aurich JC (2024) Evaluating the environmental impact of additive manufacturing: A methodology to determine the environmental impact of parts manufactured by high-speed laser directed energy deposition. *Procedia CIRP* 122:91–96. <https://doi.org/10.1016/j.procir.2024.01.013>. **(31st CIRP Conference on Life Cycle Engineering)**
94. Ehmsen S, Yi L, Glatt M, Aurich JC (2023) Evaluating the environmental impact of high-speed laser directed energy deposition: a life cycle assessment. *Procedia CIRP* 120:1606–1611. <https://doi.org/10.1016/j.procir.2023.12.003>. **(56th CIRP International Conference on Manufacturing Systems 2023)**
95. Ermini C, Visintin F, Boffelli A (2024) Understanding supply chain orchestration mechanisms to achieve sustainability-oriented innovation in the textile and fashion industry. *Sustain Prod Consum* 49:415–430. <https://doi.org/10.1016/j.spc.2024.07.008>
96. Esguerra JL, Carlsson A, Johansson J, Anderberg S (2024) Characterization, recyclability, and significance of plastic packaging in mixed municipal solid waste for achieving recycling targets in a Swedish city. *J Clean Prod* 468:143014. <https://doi.org/10.1016/j.jclepro.2024.143014>
97. Fareed AG, De Felice F, Forcina A, Petrillo A (2024) Role and applications of advanced digital technologies in achieving sustainability in multimodal logistics operations: a systematic literature review. *Sustain Futures* 8:100278. <https://doi.org/10.1016/j.sfr.2024.100278>
98. Fiorello M, Gladysz B, Corti D, Wybraniak-Kujawa M, Ejsmont K, Sorlini M (2023) Towards a smart lean green production paradigm to improve operational performance. *J Clean Prod* 413:137418. <https://doi.org/10.1016/j.jclepro.2023.137418>
99. Franke M, Deng Q, Hribernik KA, Thoben K-D, Ciaccio G (2024) Towards a Service Marketplace to Empower Circular Economy Transition: An Example Application in the Supply Chain of Textile Industry. In: Thürer M, Riedel R, von Cieminski G, Romero D (eds) *Advances in Production Management Systems. Production Management Systems for Volatile, Uncertain, Complex, and Ambiguous Environments*. Springer Nature Switzerland, Cham, pp 235–249. https://doi.org/10.1007/978-3-031-71622-5_16
100. Franzè C, Pesce D, Kalverkamp M, Pehlken A (2023) “Scale without mass”: a decision-making tool for scaling remanufacturing practices in the white goods industry. *J Clean Prod* 417:138078. <https://doi.org/10.1016/j.jclepro.2023.138078>
101. Freitas J, Sousa C, Pereira C, Pinto P, Ferreira R, Diogo R (2024) Industrial Data Sharing Ecosystems: An Innovative Value Chain Traceability Platform Based in Data Spaces. In: Rocha Á, Adeli H, Dzemmyda G, Moreira F, Poniszewska-Marañá A (eds) *Good Practices and New Perspectives in Information Systems and Technologies*. Springer Nature Switzerland, Cham, pp 423–432. https://doi.org/10.1007/978-3-031-60221-4_40
102. Gallina V, Gal B, Szaller Á, Bachlechner D, Ilie-Zudor E, Sihn W (2023) Reducing Remanufacturing Uncertainties with the Digital Product Passport. In: Kohl H, Seliger G, Dietrich F (eds) *Manufacturing Driving Circular Economy*. Springer International Publishing, Cham, pp 60–67. https://doi.org/10.1007/978-3-031-28839-5_7
103. Gallina V, Steinwender A, Zudor E, Preuveneers D, Schlund S (2024) Business model development concept for SMEs in the era of twin transition. *Procedia Comput Sci* 232:523–532. <https://doi.org/10.1016/j.procs.2024.01.052>
104. Geist H, Balle F (2024) A circularity engineering focused empirical status quo analysis of automotive remanufacturing processes. *Resour Conserv Recycl* 201:107328. <https://doi.org/10.1016/j.resconrec.2023.107328>
105. Gianvincenzi M, Marconi M, Mosconi EM, Tola F (2024) A Standardized Data Model for the Battery Passport: Paving the Way for Sustainable Battery Management. *Procedia CIRP* 122:103–108. <https://doi.org/10.1016/j.procir.2024.01.014>

106. Giovanardi M, Konstantinou T, Pollo R, Klein T (2023) Internet of things for building façade traceability: a theoretical framework to enable circular economy through life-cycle information flows. *J Clean Prod* 382:135261. <https://doi.org/10.1016/j.jclepro.2022.135261>
107. Gleich K, Behrendt S, Hörger M, Benfer M, Lanza G (2024) An Asset Administration Shell-Based Digital Product Passport as a Gaia-X Service. *Procedia CIRP* 127:224–229. <https://doi.org/10.1016/j.procir.2024.07.039>
108. Gómez-Gil M, Sesana MM, Salvalai G, Espinosa-Fernández A, López-Mesa B (2023) The digital building logbook as a gateway linked to existing national data sources: the cases of Spain and Italy. *J Build Eng* 63:105461. <https://doi.org/10.1016/j.jobte.2022.105461>
109. Gupta A, Jangra S (2024) Green human resource management and work engagement: linking HRM performance attributions. *Sustain Futur* 7:100174. <https://doi.org/10.1016/j.sfsr.2024.100174>
110. Haswell F, Edelenbosch OY, Piscicelli L, van Vuuren DP (2024) The geography of circularity missions: a cross-country comparison of circular economy policy approaches in the Global North and Global South. *Environ Innov Soc Transitions* 52:100883. <https://doi.org/10.1016/j.eist.2024.100883>
111. Haupt J, Cerdas F, Herrmann C (2024) Derivation of requirements for life cycle assessment-related information to be integrated in digital battery passports. *Procedia CIRP* 122:300–305. <https://doi.org/10.1016/j.procir.2024.01.044>
112. Hedberg A, Šipka S (2021) Toward a circular economy: the role of digitalization. *One Earth* 4:783–785. <https://doi.org/10.1016/j.oneear.2021.05.020>
113. Heeß P, Rockstuhl J, Körner M-F, Strüker J (2024) Enhancing trust in global supply chains: conceptualizing Digital Product Passports for a low-carbon hydrogen market. *Electron Mark* 34:10. <https://doi.org/10.1007/s12525-024-00690-7>
114. Horn S, Mölsä KM, Sorvari J, Tuovila H, Heikkilä P (2023) Environmental sustainability assessment of a polyester T-shirt – comparison of circularity strategies. *Sci Total Environ* 884:163821. <https://doi.org/10.1016/j.scitotenv.2023.163821>
115. Hörner Bussolo G, Bressanelli G, Visintin F, Saccani N (2024) Toward the European Union 2030 Strategy for Textiles: A Review. In: Thüerer M, Riedel R, von Cieminski G, Romero D (eds) *Advances in Production Management Systems. Production Management Systems for Volatile, Uncertain, Complex, and Ambiguous Environments*. Springer Nature Switzerland, Cham, pp 382–397. https://doi.org/10.1007/978-3-031-71622-5_26
116. Hsu W-T, Petavratzi E, Zils M, Einarsson S, Morasae EK, Lysaght O, Hopkinson P (2024) Mapping the flows and stocks of permanent magnets rare earth elements for powering a circular economy in the UK. *Sustain Prod Consum* 47:37–46. <https://doi.org/10.1016/j.spc.2024.03.027>
117. Hulea M, Miron R, Muresan V (2024) Digital Product Passport implementation based on multi-blockchain approach with decentralized identifier provider. *Appl Sci* 14:4874. <https://doi.org/10.3390/app14114874>
118. Jansen, M., Blomqvist, E., Keskinäkö, R., Li, H., Lindecrantz, M., Wannerberg, K., Pomp, A., Meisen, T., Berg, H., n.d. *Modelling Digital Product Passports for the Circular Economy*.
119. Jensen SF, Kristensen JH, Christensen A, Waehrens BV (2024) An ecosystem orchestration framework for the design of digital product passports in a circular economy. *Bus Strat Env* 33:7100–7117. <https://doi.org/10.1002/bse.3868>
120. Juhlin, P., Othen, R., Zehnpfund, A., Song, C., Schlake, J., Schmeiser, A., Kratzer, K., Sejdija, J., Kayser, P., 2024. Open Reference Architecture for Sustainable Papermaking based on Industrial Interoperability Standards and Cloud-Native Technologies, in: 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA). Presented at the 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA), pp. 1–4. <https://doi.org/10.1109/ETFA61755.2024.10711031>
121. Jung, J., Jeong, E., Song, B., Watson, K., 2024. Generate Digital Product Passports by Integrating Production Data into the Asset Administration Shell, in: 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA). Presented at the 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA), pp. 1–4. <https://doi.org/10.1109/ETFA61755.2024.10711027>
122. Kalaboukas K, Kiritsis D, Arampatzis G (2023) Governance framework for autonomous and cognitive digital twins in agile supply chains. *Comput Ind* 146:103857. <https://doi.org/10.1016/j.compind.2023.103857>
123. Kaltenegger J, Frandsen KM, Petrova E (2024) A data management perspective on building material classification: a systematic review. *J Build Eng* 92:109172. <https://doi.org/10.1016/j.jobte.2024.109172>
124. Kebede R, Moscati A, Tan H, Johansson P (2024) A modular ontology modeling approach to developing digital product passports to promote circular economy in the built environment. *Sustain Prod Consum* 48:248–268. <https://doi.org/10.1016/j.spc.2024.05.007>

125. Kessler T, Winnewisser P, Groche P (2023) Curvature measurement in profile bending - proposal and analysis of two application oriented measurement concepts. *Measurement* 217:112956. <https://doi.org/10.1016/j.measurement.2023.112956>
126. Klein P, Preisig HA, Hagelien TF, Konchakova N (2024) Optimisation of a Production Chain for Active Corrosion Protection via Digitalisation. In: Manenti F, Reklaitis GV (eds) *Computer Aided Chemical Engineering, 34 European Symposium on Computer Aided Process Engineering / 15 International Symposium on Process Systems Engineering*. Elsevier, pp 2977–2982. <https://doi.org/10.1016/B978-0-443-28824-1.50497-X>
127. Kurniawan TA, Meidiana C, Goh HH, Zhang D, Othman MHD, Aziz F, Anouzla A, Sarangi PK, Pasarihu B, Ali I (2024) Unlocking synergies between waste management and climate change mitigation to accelerate decarbonization through circular-economy digitalization in Indonesia. *Sustain Prod Consum* 46:522–542. <https://doi.org/10.1016/j.spc.2024.03.011>
128. Kuzminykh A, Granja J, Parente M, Azenha M (2024) Promoting circularity of construction materials through demolition digitalisation at the preparation stage: information requirements and openBIM-based technological implementation. *Adv Eng Inform* 62:102755. <https://doi.org/10.1016/j.aei.2024.102755>
129. Langley DJ, Rosca E, Angelopoulos M, Kamminga O, Hooijer C (2023) Orchestrating a smart circular economy: guiding principles for digital product passports. *J Bus Res* 169:114259. <https://doi.org/10.1016/j.jbusres.2023.114259>
130. Liu A, Lu S, Tao F, Anwer N (2024) Integration of data science with product design towards data-driven design. *CIRP Ann* 73:509–532. <https://doi.org/10.1016/j.cirp.2024.06.003>
131. Liu H, Wu V, Weill M, Li S, Yang X, Politis DJ, Wang L (2024) Developing physics-informed filters to align unattributed fragmental manufacturing data against a digital characteristics space (DCS). *J Manuf Syst* 77:18–25. <https://doi.org/10.1016/j.jmsy.2024.09.002>
132. Liu H, Yang X, Weill M, Li S, Wu V, Politis DJ, Shi H, Zhang Z, Wang L (2024) Unlocking inherent values of manufacturing metadata through digital characteristics (DC) alignment. *Comput Ind* 163:104148. <https://doi.org/10.1016/j.compind.2024.104148>
133. Mallalieu A, Isaksson Hallstedt S, Isaksson O, Watz M, Almefelt L (2024) Barriers and enablers for the adoption of sustainable design practices using new design methods – accelerating the sustainability transformation in the manufacturing industry. *Sustain Prod Consum* 51:137–158. <https://doi.org/10.1016/j.spc.2024.08.023>
134. Mallick PK, Salling KB, Pigosso DCA, McAloone TC (2024) Designing and operationalising extended producer responsibility under the EU Green Deal. *Environ Challenges* 16:100977. <https://doi.org/10.1016/j.envc.2024.100977>
135. Mallik AK (2023) The future of the technology-based manufacturing in the European Union. *Results Eng* 19:101356. <https://doi.org/10.1016/j.rineng.2023.101356>
136. Mêda, P., Calvetti, D., Hjelseth, E., Sousa, H., 2024. Digital Product Passports in Construction – Barriers and Opportunities at People, Process and Technology dimensions. Presented at the 2024 European Conference on Computing in Construction. <https://doi.org/10.35490/EC3.2024.233>
137. Medaglia R, Rukanova B, Zhang Z (2024) Digital government and the circular economy transition: an analytical framework and a research agenda. *Gov Inf Q* 41:101904. <https://doi.org/10.1016/j.giq.2023.101904>
138. Mendoza JMF, Pigosso DCA (2023) How ready is the wind energy industry for the circular economy? *Sustain Prod Consum* 43:62–76. <https://doi.org/10.1016/j.spc.2023.10.016>
139. Menny T, Le Guirrec S, De Wolf C (2024) The butterfly matchmaking model for circular construction: Towards a digital matchmaking platform tailored to French policy. *Sustain Prod Consum* 49:130–143. <https://doi.org/10.1016/j.spc.2024.06.011>
140. Miron R, Hulea M (2024) Digital Product Passport Implementation Based on Hyperledger Fabric Technology. In: Machado J, Soares F, Trojanowska J, Ivanov V, Antosz K, Cagaňová D, Manupati VK, Pereira A (eds) *Innovations in Industrial Engineering III*. Springer Nature Switzerland, Cham, pp 51–62. https://doi.org/10.1007/978-3-031-61582-5_5
141. Morganti L, Esnarrizaga PE, Pracucci A, Zaffagnini T, Cortes VG, Rudenå A, Brunklaus B, Larraz JA (2024) Data-driven and LCA-based Framework for environmental and circular assessment of Modular Curtain Walls. *Journal of Facade Design and Engineering* 12:9–42. <https://doi.org/10.47982/jfde.2024.305>
142. Möslinger M, Ulpiani G, Vettres N (2023) Circular economy and waste management to empower a climate-neutral urban future. *J Clean Prod* 421:138454. <https://doi.org/10.1016/j.jclepro.2023.138454>
143. Neri A, Butturi MA, Sauer HL, Lolli F, Gamberini R, Sellitto MA (2024) Distributed Ledger Technology selection for Digital Battery Passport: A BWM-TOPSIS approach*. *IFAC-PapersOnLine* 58:480–485. <https://doi.org/10.1016/j.ifacol.2024.09.258>. (18th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2024)

144. Nissinen A, Seppälä J, Heinonen T (2022) Make carbon footprints available – And it is not just one value. *Clean Logist Supply Chain* 3:100023. <https://doi.org/10.1016/j.clsn.2021.100023>
145. Nowacki, S., Sisik, G.M., Angelopoulos, C.M., 2023. Digital Product Passports: Use Cases Framework and Technical Architecture Using DLT and Smart Contracts, in: 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT). Presented at the 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), pp. 373–380. <https://doi.org/10.1109/DCOSS-IoT58021.2023.00067>
146. Nyffenegger R, Boukhatmi A, Radavičius T, Tvaronavičienė M (2024) How circular is the European photovoltaic industry? Practical insights on current circular economy barriers, enablers, and goals. *J Clean Prod* 448:141376. <https://doi.org/10.1016/j.jclepro.2024.141376>
147. Olsthoorn M, Schleich J, Guetlein M-C, Durand A, Faure C (2023) Beyond energy efficiency: Do consumers care about life-cycle properties of household appliances? *Energy Policy* 174:113430. <https://doi.org/10.1016/j.enpol.2023.113430>
148. Ospital P, Masson D, Beler C, Legardeur J (2023) Toward product transparency: communicating traceability information to consumers. *Int J Fash Des Technol Educ* 16:186–197. <https://doi.org/10.1080/17543266.2022.2142677>
149. Ott J, Schoegl J-P, Baumgartner RJ (2024) End of life focused data model for a digital battery passport. *Procedia CIRP*, 31st CIRP Conference on Life Cycle Engineering 122:277–281. <https://doi.org/10.1016/j.procir.2024.01.040>
150. Pachimuthu D, Pinzone M, Taisch M (2024) Assessing zero-defect manufacturing maturity: a review of the state of the art. *IFAC-PapersOnLine* 58:899–904. <https://doi.org/10.1016/j.ifacol.2024.09.167>
151. Palm, D., Kuenster, N., Dorka, F., Buecheler, M., 2024. Architectures And Systems For Identifying Assets In Circular Supply Chains Using Digital Product Passports And The Asset Administration Shell 898–905. <https://doi.org/10.15488/17776>
152. Panza L, Bruno G, Lombardi F (2023) Integrating absolute sustainability and social sustainability in the digital product passport to promote Industry 5.0. *Sustainability* 15:12552. <https://doi.org/10.3390/su151612552>
153. Papacharalampopoulos A, Toupalis M, Lagios P, Karagianni OM, Stavropoulos P (2024) Acquiring feedback into the manufacturing process design from Industrial policies. *Procedia CIRP* 128:579–584. <https://doi.org/10.1016/j.procir.2024.07.054>
154. Papamichael I, Voukkali I, Economou F, Liscio MC, Sospiro P, Naddeo V, Zorpas AA (2024) Investigation of customer behavior regarding circular fashion. *Sustain Chem Pharm* 41:101675. <https://doi.org/10.1016/j.scp.2024.101675>
155. Papamichael I, Voukkali I, Loizia P, Rodriguez-Espinosa T, Pedreño JN, Zorpas AA (2023) Textile waste in the concept of circularity. *Sustain Chem Pharm* 32:100993. <https://doi.org/10.1016/j.scp.2023.100993>
156. Patrucco AS, Trabucchi D, Buganza T, Muzellec L, Ronteau S (2024) Technology-enabled multi-sided platforms in B2B relationships: a critical analysis and directions for future research. *Ind Mark Manage* 122:A2–A11. <https://doi.org/10.1016/j.indmarman.2024.08.012>
157. Pehlken A, R MFD, Dawel L, Meyer O (2024) Digital Twins: Enhancing Circular Economy through Digital Tools. *Procedia CIRP* 122:563–568. <https://doi.org/10.1016/j.procir.2024.01.082>
158. Peiseler L, Wood V, Schmidt TS (2023) Reducing the carbon footprint of lithium-ion batteries, what's next? *Energy* 1:100017. <https://doi.org/10.1016/j.nxener.2023.100017>
159. Petelin E (2024) Security priorities in circular economy: a conceptual review. *Sustain Prod Consum* 47:655–669. <https://doi.org/10.1016/j.spc.2024.05.004>
160. Picatoste A, Schulz-Mönnhoff M, Niero M, Justel D, Mendoza JMF (2024) Comparing the circularity and life cycle environmental performance of batteries for electric vehicles. *Resour Conserv Recycl* 210:107833. <https://doi.org/10.1016/j.resconrec.2024.107833>
161. Pinyol Alberich J, Pansera M, Hartley S (2023) Understanding the EU's circular economy policies through futures of circularity. *J Clean Prod* 385:135723. <https://doi.org/10.1016/j.jclepro.2022.135723>
162. Pohlmann, A., Popowicz, M., Schöggel, J.-P., Bachler, J., Keler, J., Baumgartner, R.J., 2024a. Conceptualization of a digital product passport to enable circular and sustainable automotive value chains – the combustion engine use case. *Procedia CIRP*, 31st CIRP Conference on Life Cycle Engineering 122, 169–174. <https://doi.org/10.1016/j.procir.2024.01.025>
163. Pomp, A., Jansen, M., Berg, H., Meisen, T., 2023. SPACE_DS: Towards a Circular Economy Data Space, in: Companion Proceedings of the ACM Web Conference 2023, WWW '23 Companion. Association for Computing Machinery, New York, NY, USA, pp. 1500–1501. <https://doi.org/10.1145/3543873.3587685>
164. Popowicz, M., Pohlmann, A., Schöggel, J.-P., Baumgartner, R.J., 2024. Circular and sustainable battery design – The case of digital product passports as information providers during the design phase. *Procedia CIRP*, 34th CIRP Design Conference 128, 502–507. <https://doi.org/10.1016/j.procir.2024.03.031>

165. Pourjafarian, M., Plociennik, C., Rimaz, M.H., Stein, P., Vogelgesang, M., Li, C., Knetsch, S., Bergweiler, S., Ruskowski, M., 2023. A Multi-Stakeholder Digital Product Passport Based on the Asset Administration Shell, in: 2023 IEEE 28th International Conference on Emerging Technologies and Factory Automation (ETFA). Presented at the 2023 IEEE 28th International Conference on Emerging Technologies and Factory Automation (ETFA), pp. 1–8. <https://doi.org/10.1109/ETFA54631.2023.10275715>
166. Pristerà G, Tonini D, Lamperti Tornaghi M, Caro D, Sala S (2024) Taxonomy of design for deconstruction options to enable circular economy in buildings. *Resour Environ Sustain* 15:100153. <https://doi.org/10.1016/j.resenv.2024.100153>
167. Psarommatis, F., Konstantinidis, F., Azamfirei, V., May, G., 2024b. Identification of the benefits from the use of Digital Product Passport in a value chain and single organizations. IFAC-Pap., 18th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2024 58, 301–306. <https://doi.org/10.1016/j.ifacol.2024.09.199>
168. Puglia M, Parker L, Clube RKM, Demirel P, Aurisicchio M (2024) The circular policy canvas: mapping the European Union's policies for a sustainable fashion textiles industry. *Resour Conserv Recycl* 204:107459. <https://doi.org/10.1016/j.resconrec.2024.107459>
169. Pulikottil, T., Boix Rodríguez, N., Peeters, J.R., 2024. Robotic ease of Disassembly Metric (Re-DiM) for human robot cooperative disassembly: A case study for a vacuum cleaner. *Procedia CIRP*, 31st CIRP Conference on Life Cycle Engineering 122, 677–682. <https://doi.org/10.1016/j.procir.2024.02.021>
170. Quinteros-Condorety AR, Laukkanen M, Kainiemi L, Pinto SM, Lourenço EJ, Oliveira L, Albareda L, Barbiellini B (2025) Conceptual model for extending electric vehicle battery lifetime. *Resour Conserv Recycl* 212:107943. <https://doi.org/10.1016/j.resconrec.2024.107943>
171. Redeker, M., Simikin, S., Marek, F., 2024. Towards a Stateless Cloud Native and Cloud Agnostic Digital Twin Platform for the Digital Product Passport: Out-of-the-box, Reliable, and Scaling, in: 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA). Presented at the 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA), pp. 1–8. <https://doi.org/10.1109/ETFA61755.2024.10710875>
172. Reich, R.H., Alaerts, L., Van Acker, K., 2024. Towards a service-oriented architecture for information systems in the circular economy. *Procedia CIRP*, 31st CIRP Conference on Life Cycle Engineering 122, 653–658. <https://doi.org/10.1016/j.procir.2024.02.020>
173. Reich, R.H., Ayan, J., Alaerts, L., Van Acker, K., 2023. Defining the goals of Product Passports by circular product strategies. *Procedia CIRP*, 30th CIRP Life Cycle Engineering Conference 116, 257–262. <https://doi.org/10.1016/j.procir.2023.02.044>
174. Rejeb, H.B., Muxika, E., Ghadimi, P., 2024. Summer School on Circular Economy for Sustainable Manufacturing: A Case Study and Lessons Learned. *Procedia CIRP*, 31st CIRP Conference on Life Cycle Engineering 122, 952–957. <https://doi.org/10.1016/j.procir.2024.01.129>
175. Ribeiro da Silva E, Lohmer J, Rohla M, Angelis J (2023) Unleashing the circular economy in the electric vehicle battery supply chain: a case study on data sharing and blockchain potential. *Resour Conserv Recycl* 193:106969. <https://doi.org/10.1016/j.resconrec.2023.106969>
176. Riemens J, Lemieux A-A, Lussagne M, Lamouri S (2023) Apprehending traceability implementation in support of sustainable value chains: a novel analysis framework for the fashion industry. *J Clean Prod* 414:137501. <https://doi.org/10.1016/j.jclepro.2023.137501>
177. Rimaz, M.H., Plociennik, C., Ruskowski, M., n.d. Semantic Asset Administration Shell for Circular Economy.
178. Rizos V, Urban P (2024) Barriers and policy challenges in developing circularity approaches in the EU battery sector: an assessment. *Resour Conserv Recycl* 209:107800. <https://doi.org/10.1016/j.resconrec.2024.107800>
179. Rosado Da Cruz, A., Silva, P., Serra, S., Rodrigues, R., Pinto, P., Cruz, E., 2024. Data Quality Assessment for the Textile and Clothing Value-Chain Digital Product Passport, in: Proceedings of the 26th International Conference on Enterprise Information Systems. Presented at the 26th International Conference on Enterprise Information Systems, SCITEPRESS - Science and Technology Publications, Angers, France, pp. 288–295. <https://doi.org/10.5220/0012732900003690>
180. Rukanova, B., Lennartz, J., Agahari, W., Ubacht, J., Tan, Y.H., Rietveld, E., Chirvasuta, T., n.d. What to Monitor? In Search for High-level Concepts for Circular Economy and Sustainability Monitoring.
181. Rukanova, B., Ubacht, J., Tan, Y.-H., 2024. Border Crossing and Circular Economy Monitoring in a Global Context: Challenges and Opportunities, in: Proceedings of the 25th Annual International Conference on Digital Government Research, Dg.o '24. Association for Computing Machinery, New York, NY, USA, pp. 331–340. <https://doi.org/10.1145/3657054.3657196>
182. Rumetshofer T, Straka K, Fischer J (2024) How the digital product passport can lead the plastics industry towards a circular economy, a case study from bottle caps to frisbees. *Polymers* 16:1420. <https://doi.org/10.3390/polym16101420>

183. Sá M, Guise C, Costa F, Rodrigues P, Oliveira J, Barros A, Silva R, Alves T, Santos M, da Cruz AMR, Alves M, Silva CJ (2024) STVgoDigital: A Digital Product Passport Solution. In: Guarda T, Portela F, Diaz-Nafria JM (eds) *Advanced Research in Technologies, Information, Innovation and Sustainability*. Springer Nature Switzerland, Cham, pp 368–376. https://doi.org/10.1007/978-3-031-48930-3_28
184. Sakao T, Bocken N, Nasr N, Umeda Y (2024) Implementing circular economy activities in manufacturing for environmental sustainability. *CIRP Ann* 73:457–481. <https://doi.org/10.1016/j.cirp.2024.06.002>
185. Schauman S, Greene S, Korkman O (2023) Sufficiency and the dematerialization of fashion: how digital substitutes are creating new market opportunities. *Bus Horiz* 66:741–751. <https://doi.org/10.1016/j.bushor.2023.03.003>
186. Schögl J-P, Baumgartner RJ, O'Reilly CJ, Bouchouireb H, Göransson P (2024) Barriers to sustainable and circular product design – a theoretical and empirical prioritisation in the European automotive industry. *J Clean Prod* 434:140250. <https://doi.org/10.1016/j.jclepro.2023.140250>
187. Schögl J-P, Rusch M, Stumpf L, Baumgartner RJ (2023) Implementation of digital technologies for a circular economy and sustainability management in the manufacturing sector. *Sustain Prod Consum* 35:401–420. <https://doi.org/10.1016/j.spc.2022.11.012>
188. Schubert J, Ebert T, Schlegel M-C, Rödig L, Jepsen D, Memelink R, Hauschke F (2024) The front-runner approach, facilitating progressive product policy by using information from EU product databases. *Energies* 17:504. <https://doi.org/10.3390/en17020504>
189. Siska, V., 2023. Building a Sustainable Battery Supply Chain with Digital Battery Passports. <https://doi.org/10.35011/IDIMT-2023-347>
190. Sousa C, Ferreira R, Pinto P, Pereira C, Rebelo R (2024) Digital Product Passport Architecture for Boosting Circularity in Footwear Industry. *Procedia Comput Sci* 239:1560–1567. <https://doi.org/10.1016/j.procs.2024.06.331>. (CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN - International Conference on Project MANAGEMENT / HCist - International Conference on Health and Social Care Information Systems and Technologies 2023)
191. Steiner B, Münch C (2024) Leveraging digital data spaces in purchasing and supply management: paving the way to the circular economy exemplified by Catena-X. *J Purch Supply Manag* 30:100951. <https://doi.org/10.1016/j.pursup.2024.100951>
192. Steinwender A, Gallina V, Litvyak O, Lampoltshammer TJ, Bachlechner D, Schlund S (2024) From analogue to digital product passports in the furniture industry. *IFAC-PapersOnLine* 58:229–234. <https://doi.org/10.1016/j.ifacol.2024.09.175>. (18th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2024)
193. Stridsland T, Bjørnstad W, Vigen K, Østergaard KL, Sanderson H (2023) No-one left behind: an open access approach to estimating the carbon footprint of a Danish clothing company. *J Clean Prod* 426:139126. <https://doi.org/10.1016/j.jclepro.2023.139126>
194. Stucki M, Meierhofer J, Gal B, Gallina V, Eisl S (2024) Data driven value creation in industrial services including remanufacturing. *Procedia Comput Sci* 232:2240–2248. <https://doi.org/10.1016/j.procs.2024.02.043>. (5th International Conference on Industry 4.0 and Smart Manufacturing (ISM 2023))
195. Tarashkar M, Qureshi S, Wang Z, Rahimi A (2024) Public perceptions towards urban horticulture in front-yard greenhouses: unveiling ecosystem services and practices for sustainable and resilient city. *Sustain Futur* 7:100205. <https://doi.org/10.1016/j.sfr.2024.100205>
196. Tarroja B, Schoenung JM, Ogunseit O, Kendall A, Qiu Y, Malloy T, Peters J, Cha JM, Mulvaney D, Heidrich O, Baumann M (2024) Overcoming barriers to improved decision-making for battery deployment in the clean energy transition. *iScience* 27:109898. <https://doi.org/10.1016/j.isci.2024.109898>
197. Tell AW (2024) Exploring the Information Flow and the Grounding of Digital Product Passports Using the Work-Oriented Approach, an Industrial Case Study. In: Řepa V, Matulevičius R, Laurenzi E (eds) *Perspectives in Business Informatics Research*. Springer Nature Switzerland, Cham, pp 242–257. https://doi.org/10.1007/978-3-031-71333-0_16
198. Thomassen G, Peeters E, Van Hee N, Noëth E, Du Bois E, Boone L, Compernelle T (2024) The environmental impacts of reusable rice packaging: an extended comparative life cycle assessment. *Sustain Prod Consum* 45:333–347. <https://doi.org/10.1016/j.spc.2024.01.014>
199. Timms, P.D., King, M.R.N., 2023. Complexity in the delivery of product passports: a system of systems approach to passport lifecycles, in: 2023 18th Annual System of Systems Engineering Conference (SoSe). Presented at the 2023 18th Annual System of Systems Engineering Conference (SoSe), pp. 1–8. <https://doi.org/10.1109/SoSE59841.2023.10178575>
200. Tolentino-Zondervan F, DiVito L (2024) Sustainability performance of Dutch firms and the role of digitalization: the case of textile and apparel industry. *J Clean Prod* 459:142573. <https://doi.org/10.1016/j.jclepro.2024.142573>
201. Tolio TAM, Monostori L, Váncza J, Sauer O (2023) Platform-based manufacturing. *CIRP Ann* 72:697–723. <https://doi.org/10.1016/j.cirp.2023.04.091>

202. Tong X, Wang T, Li J, Wang X (2024) Extended producer responsibility to reconstruct the circular value chain. *Circ Econ* 3:100076. <https://doi.org/10.1016/j.ccc.2024.100076>
203. Traverso M, Mankaa R, Concetta Pedalá M, Covais A (2024) Social hotspot analysis of the e-waste sector in Ghana and Nigeria. *Waste Manag* 183:271–277. <https://doi.org/10.1016/j.wasman.2024.05.023>
204. Tsunazawa M, Sugiyama K, Watanabe T, Kishita Y, Umeda Y (2024) Framework for a circular economy business enabled by digital platforms: A review. *Procedia CIRP* 122:783–788. <https://doi.org/10.1016/j.procir.2024.01.109>. (31st CIRP Conference on Life Cycle Engineering)
205. van Capelleveen G, Vegter D, Olthaar M, van Hillegersberg J (2023) The anatomy of a passport for the circular economy: a conceptual definition, vision and structured literature review. *Resources, Conservation & Recycling Advances* 17:200131. <https://doi.org/10.1016/j.rcradv.2023.200131>
206. van Engelenburg, S., Rukanova, B., Ubacht, J., Tan, S.L., Tan, Y.-H., Janssen, M., 2022. From requirements to a research agenda for governments governing reuse of critical raw materials in the circular economy, in: Proceedings of the 23rd Annual International Conference on Digital Government Research, Dg.o '22. Association for Computing Machinery, New York, NY, USA, pp. 62–67. <https://doi.org/10.1145/3543434.3543645>
207. Van Opstal W, Smeets A (2023) When do circular business models resolve barriers to residential solar PV adoption? Evidence from survey data in Flanders. *Energy Policy* 182:113761. <https://doi.org/10.1016/j.enpol.2023.113761>
208. Vellesalu A, Chkanikova O, Hjelmgren D, Salomonson N (2023) Institutional re-configuration and value co-creation in circular product development: a service ecosystem perspective in the textile and apparel industry. *J Clean Prod* 414:137682. <https://doi.org/10.1016/j.jclepro.2023.137682>
209. Vomborg A, de Haan E, Etienne Fabian N, Broekhuizen T (2024) Digital knowledge engineering for strategy development. *J Bus Res* 177:114632. <https://doi.org/10.1016/j.jbusres.2024.114632>
210. Voukkali I, Papamichael I, Loizia P, Economou F, Stylianou M, Naddeo V, Zorpas AA (2024) Fashioning the future: Green chemistry and engineering innovations in biofashion. *Chem Eng J* 497:155039. <https://doi.org/10.1016/j.cej.2024.155039>
211. Voulgaridis, K., Lagkas, T., Karampatzakis, D., Argyriou, V., Sarigiannidis, P., 2023. Realizing Digital Product Passports with Crowdsourcing Principles: The Case of Sustainable Smart Grids, in: 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT). Presented at the 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), pp. 381–388. <https://doi.org/10.1109/DCOSS-IoT58021.2023.00068>
212. Wagner, E., Kegler, S., Gallina, V., Aigner, T., Koller, J., 2024a. How Industrial Data Meets Information Requirements of the Digital Product Passport – The Case of Remanufacturing Data, in: 2024 Electronics Goes Green 2024+ (EGG). Presented at the 2024 Electronics Goes Green 2024+ (EGG), pp. 1–7. <https://doi.org/10.23919/EGG62010.2024.10631243>
213. Weber S, Forte S, Dickopf T, Kirsch L, Apostolov C (2024) Supporting Sustainable Product Design with Engineering Data Management Capabilities. *Procedia CIRP* 122:271–276. <https://doi.org/10.1016/j.procir.2024.01.039>. (31st CIRP Conference on Life Cycle Engineering)
214. Werner A, Schuseil F, Hämmerle M, Schaper S, Hölzle K (2025) Modular digital twin – an approach for generating and exploiting product sustainability information towards service-oriented business models. *Int J Prod Res* 63:992–1010. <https://doi.org/10.1080/00207543.2024.2366997>
215. Wu Y, Fu S, Long R, Islam MS, Huang E (2024) Do green information transparency and exposure always boost online sales of organic food? An evidence from China. *Electron Commer Res Appl* 65:101400. <https://doi.org/10.1016/j.elerap.2024.101400>
216. Xu W, Jia F(J), Chen L, Schoenherr T (2024) Editorial: Sustainable transition in textile and apparel industry. *J Clean Prod* 443:141081. <https://doi.org/10.1016/j.jclepro.2024.141081>
217. Zante G, Elgar CE, Hartley JM, Mukherjee R, Kettle J, Horsfall LE, Walton A, Harper GDJ, Abbott AP (2024) A toolbox for improved recycling of critical metals and materials in low-carbon technologies. *RSC Sustain* 2:320–347. <https://doi.org/10.1039/D3SU00390F>
218. Zorpas AA (2024) The hidden concept and the beauty of multiple “R” in the framework of waste strategies development reflecting to circular economy principles. *Sci Total Environ* 952:175508. <https://doi.org/10.1016/j.scitotenv.2024.175508>

Authors and Affiliations

Foivos Psarommatis^{1,2,3}  · Gokan May⁴

✉ Foivos Psarommatis
foivosp@ifi.uio.no

Gokan May
gokan.may@unf.edu

¹ SIRIUS Labs, Department of Informatics, University of Oslo, Gaustadalléen 23, Oslo B N-0373, Norway

² Zerofect GmbH, BaarAllenwinden 6319, Switzerland

³ Dept of Informatics and Telecommunications, University of Ioannina, Ioannina, Greece

⁴ School of Engineering, University of North Florida, Jacksonville, USA